

Bid-Ask Spreads, Trading Networks and the Pricing of Securitizations

by

Burton Hollifield,* Artem Neklyudov** and Chester Spatt*

December 27, 2016

* Tepper School of Business, Carnegie Mellon University.

** Department of Finance HEC, University of Lausanne and SFI.

Forthcoming in the RFS. Data was provided to us on a confidential basis by the Financial Industry Regulatory Authority to facilitate analysis of securitization markets. We thank Allen Carrion, Alié Diagne, Kathleen Hanley, Steve Joachim, Andrew Karolyi (the editor), Tim Landvoigt, Dan Li, Pamela Moulton, Ola Persson, Michael Piwowar, Uday Rajan, Norman Schürhoff, the anonymous referee, and participants at presentations at the AREUEA meetings, the Bank of Canada, the Board of Governors of the Federal Reserve Bank, Carnegie Mellon University, CKGSB, the EFA meetings, the Federal Reserve Bank of New York, the FIRS meetings, Nanyang Technical University, National University of Singapore, NBER Market Microstructure meeting, the Securities and Exchange Commission, Singapore Management University, Tsinghua University, University of Rochester, University of Waterloo, and the WFA meetings for helpful feedback.

Bid-Ask Spreads, Trading Networks and the Pricing of Securitizations

Abstract

The Financial Industry Regulatory Authority began to collect transaction data from broker-dealers in 2011 as a step towards enhancing its understanding of securitization markets. We use transaction data to document the importance of the interdealer network structure for market quality. Some dealers are relatively central in the network and trade with many dealers, while others are more peripheral. Core dealers receive relatively lower and less dispersed spreads than peripheral dealers. We develop a model in which core and peripheral dealers trade with different customer clienteles and argue that the presence of relatively sophisticated customers in securitization markets explains these facts.

Keywords: Securitization; sophisticated investors; Rule 144A; network analysis; price discovery; OTC markets; bid-ask spreads; core dealers; peripheral dealers.

JEL Classification: G12, G14, G23

1. Introduction

Relatively little is known about the pricing of securitizations, because these have traded traditionally in opaque markets. The importance of the shadow banking system, in general, and securitization, in particular, has been recognized strongly by the Financial Crisis Inquiry Commission (2011) in the aftermath of the financial crisis. To illustrate the size of securitization markets we point to the example of asset-backed securities (ABS), where investors held \$652 billion of ABS and purchased \$118 billion of new ABS in 2011, according to Securities Industry and Financial Markets Association (SIFMA). For comparison, investors purchased \$1,042 billion of new corporate bonds, according to SIFMA. Investors in securitizations are sophisticated and include hedge funds, insurance companies, pension funds, banks, asset management firms, government agencies, corporations, and mutual funds, with few retail investors directly investing in securitizations.¹

While there was a substantial move towards transparency in corporate and municipal bonds, securitization markets remained opaque and comparability little was known about these markets—even by its self-regulatory organization, the Financial Industry Regulatory Authority (FINRA). Consequently, to help better understand the securitization markets in the opaque regime, including the spreads in these markets, FINRA began to collect price information to encourage the analysis of these markets, resulting in our study and several others of securitization trading in opaque markets. For example, Bessembinder, Maxwell and Venkataraman (2013) and Friewald, Jankowitsch and Subrahmanyam (2016) describe the size and importance of the U.S. fixed-income structured product market and examine spreads in these markets. Atanasov and Merrick (2012) use the FINRA

¹ As of 2013, insurance companies and pension funds held 17% of the ABS outstanding and 35% of the commercial mortgage-backed securities, and banks held 12% of the ABS and 15% of the commercial mortgage-backed securities (Segoviano et al. 2015).

data to study the liquidity and pricing of agency mortgage-backed pass-through securities. FINRA ultimately included several securitization markets within TRACE and took a variety of steps for the previously non-disseminated asset types to move towards greater dissemination of transaction information.² Our study examines the functioning of the market prior to inclusion of securitization markets in TRACE.

We use securitization transaction data from FINRA to examine the importance of the interdealer network structure for intermediation, bid-ask spreads, information flows, and market quality. We document the core-periphery nature of interdealer trading. We also study the effects of varying customer clienteles on the dealer markets by examining the contrast between Registered instruments that require detailed disclosures in the issuance process, and Securities and Exchange Commission (SEC) Rule 144A instruments that exempt private resale of instruments to Qualified Institutional Buyers (QIBs) from these disclosure requirements. We focus our analysis upon asset-backed securities (ABS), collateralized debt obligations (CDO), commercial-mortgage-backed securities (CMBS) and collateralized mortgage obligations (CMO) instruments due to the mix of trading of 144A instruments. We benchmark 144A instruments against corresponding Registered instruments, where possible.³

Fundamentally, there are a large number of securitizations, trading is very fragmented with relatively few transactions in most individual instruments, with many instruments even not trading at all in our sample. A feature of securitization markets that differentiate it from other over-the-counter markets, such as corporate bonds or municipal bonds, is the absence of retail activity.

² In October 2011 FINRA began publishing market activity indices for securitizations of all types, in July 22, 2013 the asset-backed securities issued by Small Business Administration (SBA) were included in TRACE, and in June 1, 2015 all other securitizations except collateralized debt obligations were included in TRACE.

³ Since there are no 144A instruments in the mortgage-backed securities (MBS), securities traded on to-be-announced basis (TBA) and agency CMO securities, we do not use these in our benchmark analysis. We include the CDO category because these are largely 144A instruments.

Dealers in securitization markets have repeated interactions with their customers because these are large and sophisticated investors, making the trading network important.

In the interdealer securitization market some dealers are relatively central and trade with many dealers, while others are more peripheral. We show that the dealers' centrality in the interdealer markets is an important determinant of the terms of trade for customers; the resulting bid-ask spreads vary depending on a dealer's position in the network. We document a negative relationship between a dealer's centrality and the bid-ask spreads both for Registered and Rule 144A securitizations, which we refer to as the centrality discount. Our finding contrasts with the centrality premium found in the municipal bond markets (Li and Schürhoff 2016) and the corporate bond markets (Di Maggio, Kermani, and Song 2016).

We develop a theoretical model featuring heterogeneous customers to interpret the centrality discount in securitization markets and the centrality premium found in other markets. Several other models predict a centrality discount including Neklyudov (2013), Weller (2013), and Zhong (2014), but do not provide a natural explanation for a centrality premium. In Zhong (2014) a centrality premium can arise unconditionally, but there is a centrality discount conditional on trading volume. Üslü (2015) provides conditions for both a centrality premium and a centrality discount. A centrality premium arises when core dealers intermediate large enough trades in equilibrium. In our model customers differ in their sophistication and choose dealers based on trade execution speed and offered price. Differences in customer sophistication can arise from wealth differences or trading abilities. Relatively less sophisticated customers have weaker outside options and relatively more sophisticated customers have stronger outside options. Because peripheral dealers provide slower trade executions than core dealers, they cannot offer favorable enough pricing to attract the more sophisticated customers. Both peripheral and core dealers offer favorable enough pricing to

attract relatively less sophisticated customers. Spreads are higher with core dealers when there are few customers with strong outside options, and spreads are lower with core dealers when there are many customers with strong outside options. We provide conditions under which the peripheral dealers' distribution of bid-ask spreads is more dispersed than the core dealers' distribution of spreads. We use our model to interpret our empirical findings of a centrality discount and a higher dispersion of bid-ask spreads for peripheral dealers resulting from the presence of many relatively sophisticated customers in the securitization markets.

Some of the bid-ask spreads in the ABS, CDO, CMBS and Non-Agency CMO markets are surprisingly large—for example, the average spread for Non-Agency CMO instruments is 3.36% of the mid-quote. For ABS, CMBS and Non-Agency CMO instruments there is a volume discount with respect to the bid-ask spread—larger volume trades result in lower spreads than smaller volume trades. That larger trades obtain better prices is reminiscent of a core insight from pricing municipal bonds (Green, Hollifield, and Schürhoff 2007; Harris and Piwowar 2006) and corporate bonds (Bessembinder, Maxwell, and Venkataraman 2006; Edwards, Harris, and Piwowar 2007; Goldstein, Hotchkiss, and Sirri 2007). Bernhardt et al. (2005) show theoretically that when brokers with more valuable relationships with dealers submit larger and more frequent orders, then larger orders receive greater price improvement on average. In a competitive model with a search friction developed by Lester, Rocheteau, and Weill (2015) larger trades receive lower bid-ask spreads.

Our spread estimates control for registration status. The required disclosures are more limited for Rule 144A than for Registered securitizations. Rule 144A instruments experience a corresponding potential reduction in issuance cost and exemption from liability. The restrictions on re-trading for 144A instruments suggests these are oriented to buy-and-hold investors, which can lead to higher effective spreads and reduced liquidity. But sophisticated investors such as QIBs may have

enhanced bargaining power with dealers, leading to lower effective spreads. Indeed, empirically within some asset classes 144A securitizations have higher spreads than Registered securitizations and within other asset classes 144A securitizations have lower spreads than Registered securitizations. Historically, FINRA did not include Rule 144A instruments in TRACE. In June 2014, after our sample period ended and our paper was first circulating, FINRA changed the transaction reporting rules to be the same for TRACE-eligible Registered and Rule 144A instruments. By June 2015 all securitizations except CDO were TRACE-eligible.

We also document the intermediation chains in our sample. Several theoretical papers emphasize the role of sequential trading and intermediation chains. Viswanathan and Wang (2004) show that when dealers are relatively less informed about bond fundamentals than customers, sequential trading leads to shorter intermediation chains. Hugonnier, Lester, and Weill (2015) develop a search model in which all investors have different values for holding the asset and there are multi-round intermediation chains. Shen, Wei, and Yan (2016) show how trading costs and trading speeds affect the length of intermediation chains. Glode and Opp (2016) demonstrate how multi-round intermediation chains with informed intermediaries reduce adverse selection costs.

We apply a panel vector autoregression (VAR) approach to study the price discovery process in the network, based on Hasbrouck (2007) and the GMM estimator of Abrigo and Love (2015). Babus and Kondor (2016) develop a network model with information asymmetry. More central dealers are less exposed to adverse selection, and their trades aggregate more information than peripheral dealers' trades. While our panel VAR design is not a direct test of the Babus and Kondor (2016) framework, it is a way to study how dealer's centrality influences permanent price dynamics. We also use the panel VAR to evaluate the extent of asymmetric information for different securitization

markets. There is more adverse selection in Registered instruments than in Rule 144A instruments. Dealer centrality does not influence the price discovery process in the securitization markets.

2. A Model of Trade with Core and Peripheral Dealers and Heterogeneous Customers

Our finding of a centrality discount in securitizations differs from a centrality premium found in several other fixed-income markets, e.g. Di Maggio, Kermani, and Song (2016) for corporate bonds and Li and Schürhoff (2016) for municipal bonds. Our theoretical model provides conditions on the degree of customer heterogeneity under which either a discount or a premium can arise.

The model features customers and dealers. Heterogeneous customers choose to transact with heterogeneous dealers differing in their trade execution speeds. Core dealers use their network connections to execute customer trades faster than peripheral dealers. We present the model for the buy side, with similar results holding for the sell side. Heterogeneous buyers endogenously trade with core and peripheral dealers; the centrality discount arises when core and peripheral dealers have different trade execution technologies and serve sufficiently different types of buyers.

Risk neutral buyers discount cash flows at rate $0 < r < 1$. Once a buyer acquires the asset he or she receives a flow of dividends of 1 per unit of time and so has a holding value for the asset of $1/r$. Each buyer has reservation utility level u drawn independently from a uniform distribution on the interval $[0, U]$. Buyers with low reservation utility have weak outside options and buyers with high reservation utility have strong outside options. Differences in buyers' reservation utility levels u provide a reduced form measure of differences in wealth, demand for liquidity, risk exposures or private values for holding the asset among customers, as in Hugonnier, Lester, and Weill (2015), Neklyudov and Sambalaibat (2016), Chang and Zhang (2016), and Atkeson, Eisfeldt, and Weill (2015). An increase in U implies that there are more buyers with strong outside options. We

interpret such buyers as the more sophisticated buyers; an increase in U increases the proportion of sophisticated buyers in the market as well as an increase in buyer heterogeneity.

Dealers differ in their exogenous trade execution speeds, as in Neklyudov (2013) and Weller (2013). We denote the dealers' execution speed by λ_j for $j \in \{c, p\}$, where c denotes a core dealer and p denotes a peripheral dealer, with $\lambda_c > \lambda_p$. Dealers value the asset at $L < V_h$ so that there are gains from trade with buyers. The lowest price a dealer would sell the asset to a buyer is L .

As is standard in the search literature, a buyer's outside option u is common knowledge. For simplicity we assume that a dealer with execution speed λ_j makes ex ante a take-it-or-leave-it offer of price $P_j(u)$ to a buyer with outside option u . A buyer accepts the offer when the buyer's utility from trading is at least u . A buyer's value function of choosing dealer j with execution speed λ_j offering price $P_j(u)$ is

$$V_0(u; P_j(u)) = \frac{\lambda_j}{r + \lambda_j} \left(\frac{1}{r} - P_j(u) \right). \quad (1)$$

The buyer negotiates the price with a dealer ex ante before the trade occurs so that the negotiated price takes into account the effect of trade execution delay. Because dealers make take-it-or-leave-it offers, buyers only trade with dealers offering prices that satisfy the participation constraint with equality:

$$V_0(u; P_j(u)) = u. \quad (2)$$

Solving,

$$P_j(u) = \frac{1}{r} - \left(1 + r/\lambda_j \right) u. \quad (3)$$

A dealer with trade execution speed λ_j will only offer the price $P_j(u)$ when $P_j(u) \geq L$, otherwise the trade would not happen between the dealer with trade execution speed λ_j and the buyer with outside option u .⁴

In equation (3) the price $P_c(u)$ offered by a core dealer to a given buyer with an outside option $u > 0$ is higher than the price $P_p(u)$ offered by a peripheral dealer: Peripheral dealers make price concessions in order to compensate for their slower execution speed. The price $P_j(u)$ decreases in the buyer's outside option u . More sophisticated buyers with a stronger outside option receive lower offer prices. This is reminiscent of Duffie, Gârleanu, and Pedersen (2005, 2007), where more sophisticated investors with better search abilities receive tighter bid-ask spreads from market makers.

To describe the equilibrium define u_p as the maximum value of u such that a trade would happen between a buyer with outside option u and a peripheral dealer with trade execution speed λ_p

$$u_p = \frac{1-rL}{r(1+r/\lambda_p)}. \quad (4)$$

Similarly define u_c as the maximum value of u such that a trade would happen between a buyer with outside option u and a core dealer with trade execution speed λ_c

$$u_c = \frac{1-rL}{r(1+r/\lambda_c)} > u_p. \quad (5)$$

The equilibrium decision rule for a buyer with outside option u is

⁴ If prices arise from a Nash bargaining solution in which both core and peripheral dealers have bargaining power α , the pricing equation is:

$$P_j(u) = \alpha \frac{1}{r} + (1 - \alpha)L - \alpha \left(1 + r/\lambda_j\right) u.$$

The results in the text will remain qualitatively unaffected under such a specification as long as both core and peripheral dealers have the same bargaining power with buyers.

$$\begin{aligned} &\text{Trade with dealer } j \text{ at } P_j(u), & \text{if } u \leq u_j, \\ &\text{Do not trade with dealer } j, & \text{if } u > u_j. \end{aligned} \tag{6}$$

Figure 1 shows the decision rules for buyers. Buyers with weak outside options $u \leq u_p$ can trade with both core dealers and peripheral dealers, and are indifferent between them. When buyers are indifferent between trading with either core or peripheral dealers, they choose peripheral dealers with probability π . Buyers with better outside options $u_p < u \leq u_c$ trade only with core dealers. Buyers with the best outside options $u > u_c$ do not trade.

INSERT FIGURE 1 ABOUT HERE

Our model provides empirical predictions about the average prices received by core and peripheral dealers—a centrality premium or a centrality discount that depends on the extent of customer heterogeneity and market liquidity, and the model provides predictions about the price dispersion.

Proposition 1 (Average Prices):

1. *When $U \leq u_p$ there is a centrality premium: The average ask prices are higher for trades with core dealers than for trades with peripheral dealers.*
2. *When $u_p < U \leq u_c$ there exists a threshold value of probability $\pi^*(U)$ such that when $\pi < \pi^*$ there is a centrality premium: The average ask prices are higher for trades with core dealers than for trades with peripheral dealers. When $\pi > \pi^*$ there is a centrality discount: The average ask prices are lower for trades with core dealers than for trades with peripheral dealers. The threshold $\pi^*(U)$ is decreasing in U .*
3. *When $u_c < U$ there is a centrality discount: The average ask prices are lower for trades with core dealers than for trades with peripheral dealers.*

Proof: See Appendix A.

When the composition of buyers from core and peripheral dealers is the same, the equilibrium features a centrality premium—core dealers offer relatively larger average prices to buyers than peripheral dealers. When the composition of buyers from core dealers is sufficiently different from the composition of buyers from peripheral dealers the equilibrium features a centrality discount—core dealers offer relatively smaller average prices to buyers than peripheral dealers. In this case, equilibrium features relatively more sophisticated buyers trading with core dealers than with peripheral dealers.

We interpret a higher U as an increase of buyer heterogeneity and sophistication in the market. Proposition 1 states that markets with fewer sophisticated buyers and less buyer heterogeneity should exhibit a centrality premium and markets with more sophisticated buyers and greater buyer heterogeneity should exhibit a centrality discount. That there is a centrality discount in the markets for securitizations is consistent with sufficient customer heterogeneity and the presence of sophisticated customers.

A centrality discount can arise in several other models. For example, Neklyudov (2013) develops a search model in which dealers differ in trade execution speeds and faster dealers charge customers lower and less volatile bid-ask spreads. Weller (2013) develops a model in which dealers differ in quote updating speed, and finds that faster dealers charge customers lower and less volatile bid-ask spreads. In both Neklyudov (2013) and Weller (2013), faster dealers act as core dealers in the trading network and a centrality discount arises. Zhong (2014) predicts that risk-averse dealers with greater liquidity provision capacity become core dealers in the network. These core dealers tend to offer better prices conditional on the size of an investor's order. Unlike these other models, our model allows for either a centrality premium or a centrality discount depending on the heterogeneity

in customer sophistication. Our model helps explain why the market of securitizations can have a centrality discount while other over-the-counter markets can have a centrality premium.

In equation (5) the threshold value u_c increases as the core dealers' execution speed λ_c increases. We interpret a market with a lower core dealer execution speed as a less liquid market. Part 3 of Proposition 1 implies that for λ_c sufficiently small we expect to find a centrality discount. A centrality discount will occur for a less liquid market.

Proposition 2 (Sufficient Condition for Price Dispersion): *When $u_p \leq u_c/2$ then price dispersion for trades with core dealers is smaller than for trades with peripheral dealers, regardless of the value of probability π .*

Proof: See Appendix A.

The sufficient condition in Proposition 2, $u_p \leq u_c/2$, holds when core dealers are sufficiently faster than peripheral dealers. Proposition 2 states that the price dispersion is lower for trades with core dealers than for trades with peripheral dealers when there is enough dispersion in the dealers' trade execution speeds.

Our model assumes that all dealers have a homogeneous and exogenous reservation value to sell the asset, L . If instead we assume that core dealers have lower reservation value to sell the asset than peripheral dealers as in the models in Zhong (2014) and Malamud and Rostek (2016), then core dealers would charge lower average ask prices than peripheral dealers. A centrality discount would occur for a larger set of parameter values than when dealers have equal reservation values.

3. Our Data and Markets

Our sample contains all trading activity in ABS, CDO, CMBS and Non-Agency CMO instruments overseen by FINRA between May 16, 2011 and February 29, 2012. These data are a sequence of trade reports that provide the trade identifier, the execution timestamp and settlement date, the side of the reporting party—either the buy side or sell side, the trade volume of the trade measured in dollars of original par balance, and the trade price measured in dollars per \$100 par. The trade report allows us to determine if the trade is between a dealer and an outside customer, or between two dealers. We use the collateral type to categorize ABS instruments, we split the CDO category into Collateralized Bond Obligations (CBO), Collateralized Loan Obligations (CLO), and all other CDO instruments, and we use the tranche type to categorize CMBS and CMO instruments. We provide details on our data-cleaning procedures in Appendix B.

Rule 144A securities are considered restricted securities for a period between six months and one year after their initial offering, depending on whether the issuer is a reporting company or not. The securities can become unrestricted after the holding period elapses, can be traded outside of the QIBs market, and then would be assigned a different unrestricted CUSIP. However, many asset-backed securities due to their complex structure remain Rule 144A securities until maturity and trade only among QIBs. In our sample we track a security that changes status from Rule 144A to Registered by matching a Rule 144A CUSIP to a corresponding Registered CUSIP. We only find a few examples among ABS, CMBS and CMO instruments that change status within our sample. We provide additional details on Registered and Rule144A instruments in Appendix B.

The population of instruments is the comprehensive list of securitizations in the FINRA database that consists of all private-label and agency securitizations outstanding. Table 1 reports descriptive statistics for traded instruments. It is apparent from the trading frequencies reported in Table 1 that

securitized products do not trade frequently: For example, on average Registered ABS instruments have 19.4 trades in our sample period and Rule 144A CDO instruments have 4.5 trades in our sample period. Registered instruments are more likely to trade in our sample and tend to have more trades on average than 144A instruments, which often are purchased by QIBs as buy-and-hold investments. On average Registered instruments have 15.8 trades, and Rule 144A instruments have 6.5 trades in our sample period. Perhaps the higher frequency of trading in Registered instruments reflects that a larger number of investors can hold and trade Registered instruments than 144A instruments. The higher frequency of trading may also reflect the greater disclosure requirements for Registered instruments so that potential investors have more public information about these instruments and are more willing to trade them.

INSERT TABLE 1 ABOUT HERE

Figure 2 depicts the cumulative frequency of CUSIPs against the number of trades per CUSIP. The distribution of the number of trades for Registered instruments first-order stochastically dominates the distribution of the number of trades for Rule 144A instruments. Some CUSIPs have many more trades during our sample period than other CUSIPs within each category of instrument. For example, 23% of CDO instruments have a single trade in our sample period, while the most active CDO instrument has 130 trades.

INSERT FIGURE 2 ABOUT HERE

Dealers may possess bargaining advantages with respect to smaller-size trades. Following FINRA, we define a smaller-size trade as a trade of less than \$100,000 of current balance volume, and we define larger-size trades as all other trades. Figure 3 shows an example of transaction prices in our data by focusing on the second most actively traded ABS instrument. The figure shows the customer-buy, interdealer, and customer-sell prices for smaller-size trades in the upper panel and for

the larger-size trades in the lower panel. Bid-ask spreads are quite substantial for smaller-size transactions, more volatile than for larger-size trades, and interdealer trade prices do not always lie between the customer buy and sell trade prices.

INSERT FIGURE 3 ABOUT HERE

We compute total proportional customer bid-ask spreads by matching trades of opposite direction in our data. For intermediation chains with multiple dealers we also compute interdealer bid-ask spreads for each participating dealer at each step of the intermediation chain. The total customer bid-ask spread is a realized trading cost for the two customers taking opposite sides in the transaction. The interdealer bid-ask spread is a realized trading gain for the dealer participating in both buy and sell trades. We adjust the realized spreads for accrued interest and factor prepayments.⁵ We define smaller-size bid-ask spreads as bid-ask spreads computed from a pair of trades where either the buy trade or the sell trade is a smaller-size trade, and we define larger-size bid-ask spreads as all other spreads in our sample.

Table 2 reports the average customer bid-ask spreads for Registered and 144A instruments, and for smaller-size trades and larger-size trades. The difference in spreads between smaller-size trades and larger-size trades reported in Table 2 is striking. For almost all categories of instruments, smaller-size bid-ask spreads are more than double the larger-size bid-ask spreads. Such smaller-size bid-ask spreads are consistent with the theoretical predictions in Lester, Rocheteau, and Weill (2015) and Bernhardt et al. (2005), and may indicate that dealers exercise greater bargaining power when they intermediate smaller-size customer trades. Babus and Kondor (2016) provide an alternative explanation for why smaller-size trades have higher spreads. In their model, dealers with fewer

⁵ Few of the resulting spread observations are extreme, potentially reflecting pricing data errors. We address such observations by Winsorizing the upper 1% and lower 1% tails of the spread distributions within each of the four types of instruments and each of the two placement types.

trading partners execute smaller trades and post larger bid-ask spreads because they are exposed to larger adverse selection.

INSERT TABLE 2 ABOUT HERE

We also compare spreads across instrument types. Overall average spreads are the largest for Non-Agency CMO instruments. For Non-Agency CMO instruments 144A instruments have smaller spreads than Registered instruments, which may reflect that in the market for Non-Agency CMO instruments sophisticated investors face lower transactions costs and have greater bargaining power than unsophisticated investors.

Our findings on relative bid-ask spread sizes are consistent with other studies of securitizations based on the data from FINRA. Bessembinder, Maxwell, and Venkataraman (2013) provide similar sized trading cost estimates using a regression-based trading cost model for ABS and MBS instruments, with a focus on quantifying the effects of a potential transparency regime change. Friewald, Jankowitsch, and Subrahmanyam (2016) analyze liquidity of the markets for structured products and also demonstrate that transaction cost measures based on less detailed information are good proxies for measures based on more detailed dealer-specific information. We compare the non-retail spreads reported in Table 2 with the spreads for non-disseminated corporate bonds reported by Goldstein, Hotchkiss, and Sirri (2007) in their Table 7 for the period from July 8, 2002 until April 4, 2003. For Registered instruments the average spread of 3.320 for smaller-size trades is larger than spreads for trade sizes below 50 bonds (defined as retail-size trades in Goldstein, Hotchkiss, and Sirri (2007)) in non-disseminated corporate bonds, and the average spread of 1.191 for larger-size trades is larger than spreads for trade sizes of 51 and larger in non-disseminated corporate bonds. The spreads of similar trade sizes for 144A instruments in Table 2 tend to be comparable to the non-disseminated corporate bonds. Among securitization types, only ABS

instruments have lower average spreads than corporate bonds of similar trade sizes before dissemination.

About halfway through our sample on October 18, 2011, FINRA began to disseminate daily price index data for various collateral types of structured products. These informational releases provided the public more detailed information about aggregate valuations for various collateral types and indirectly, greater transparency about spreads and trading costs. In an Online Appendix we examine spread differences and price discovery before and after the transparency event, finding only small effects of the event relative to the underlying variability of prices and spreads. The nature of transparency provided by the index data was modest because of cross-sectional aggregation of prices combined with infrequent trading of the underlying instruments and the changing composition of the traded instruments across time.

4. Trading Networks and Intermediation Chains

We use network analysis to document the structure of the relationships among dealers. By forming relationships dealers can reduce trading frictions, such as search, opacity and informational frictions created by the nontransparent environment. Empirically, we relate the structure of the interdealer trading network and dealers' positions in the network to their trading behavior.

Our data allows us to identify individual dealers and their trades from May 16, 2011 to February 29, 2012. During the sample period 658 dealers participated at least once in interdealer trading. Of those dealers, 359 participated in ABS, 223 in CDO, 292 in CMBS, and 555 in Non-Agency CMO markets. Over the sample, an average dealer transacted \$448 million of original balance on the interdealer market in ABS, \$137 million in CDO, \$1,191 million in CMBS, and \$759 million in

Non-Agency CMO instruments. These markets are significantly interconnected and the same dealers are active in many of them.

Dealers are remarkably heterogeneous in terms of their trading activity. Figure 4 presents the Lorenz curves computed using dealers' shares of the customer volumes for ABS, CDO, CMBS and Non-Agency CMO instruments and for Registered and 144A instruments. The top 10% of dealers account for at least 85% of customer volume in all markets.

INSERT FIGURE 4 ABOUT HERE

We use the volume of interdealer trades to identify which dealers have trading relationships. We assume dealers have an underlying trading relationship in the interdealer trading network when they trade more than \$5 million over the sample period with each other. The interdealer trading network does not include pairs of dealers who never traded with each other or traded less than \$5 million over the sample period. Our results are robust to an alternative cutoff choice of \$10M. 57% of the dealers we observe in ABS have at least one trading relationship in the interdealer trading network. Similarly, 54% of the dealers in CDO, 61% in CMBS, and 54% in Non-Agency CMO markets have at least one trading relationship in the interdealer trading network. Dealers with at least one trading relationship on average trade more than 65% larger total volumes than the average dealer. Our trading network captures a relatively active part of the interdealer market.

We define a prearranged trade as a buy trade and a sell trade of the same size executed by the same dealer within 15 minutes. For example, when dealer A buys a given volume of a security from a customer and within 15 minutes resells the same volume to dealer B, we conclude that dealers A and B have a pre-established trading relationship. Otherwise, it would have been difficult for dealer A to find a trading opportunity with dealer B so quickly. Our results are robust when we keep only those pairs of interdealer trades that are part of at least one prearranged trade in constructing the

dealer network and computing the network measures. A majority of dealers with trading relationships in the interdealer network execute prearranged trades with each other.

Figure 5 depicts the interdealer network. The network has 658 dealers, each represented by circles of whose size is proportional to the number of the dealer's counterparties in the interdealer market. Relationships between pairs of counterparties are shown as lines. The figure shows the core-peripheral structure of the underlying interdealer network. To study further the interdealer network structure, we consider two network centrality measures that capture a dealer's relative position in the interdealer network. Degree centrality of dealer i is the number of direct neighbors of dealer i in the interdealer network with whom dealer i is connected. Eigenvector centrality of a dealer i is a measure of dealer i centrality that weights higher connections to more central dealers in the interdealer network. A dealer with the same number of direct neighbors would have a higher eigenvector centrality when these neighbors have more connections themselves. See Jackson (2008) for the formal definition.

INSERT FIGURE 5 ABOUT HERE

Adamic et al. (2010) study trading networks in E-mini S&P 500 futures contracts and use degree centrality and other measures to study the interactions of network metrics with transaction prices, volume and liquidity measures. Li and Schürhoff (2016) explore several centrality measures in a sample of municipal bond trades and document a significant common component in these measures.⁶ Di Maggio, Kermani, and Song (2016) use eigenvector centrality to classify dealers in the network because it contains information about all direct and indirect trading partners. Our

⁶ Li and Schürhoff (2016) construct a single aggregate proxy for a dealer's position in the interdealer market from a principal component analysis of all centrality measures. Our results are robust to the use of these measures.

empirical results in sections 5 and 6 are robust to using either degree centrality or eigenvector centrality to classify dealers.

Figure 6 depicts the degree centrality distributions of the interdealer network. Both for Registered and Rule 144A instruments the dealers' degree centrality follows a power law. As discussed in Jackson (2008), this implies that the interdealer network is a scale-free network. Dealers are more likely to be connected to dealers with greater numbers of existing connections.

INSERT FIGURE 6 ABOUT HERE

We use a matching technique to identify intermediation chains in our sample. Each chain captures the movement of a block of volume between a customer-seller and a dealer, between dealers in the interdealer network, and between a dealer and a customer-buyer.

Figure 7 depicts two examples of intermediation chains for the two most frequently traded Registered ABS in our sample. Panel A depicts an intermediation chain with one customer-seller, two dealers, and two customers-buyers. The first dealer buys \$14.95 million of original balance at the bid price from a customer and sells \$1.5 million to another customer and \$13.45 million to the second dealer. The second dealer in turn resells \$13.45 million to a customer. Both dealers hold no additional inventory after all trades are executed, and in this example the first dealer incurs a loss on the \$1.5 million transaction by selling at the lower ask price than the original bid price. On average the chain is profitable for the first dealer. Panel B depicts a different intermediation chain which involves two customers-sellers, one dealer, and one customer-buyer. We call such intermediation chains reverse-request transactions, in which the number of customers-sellers is greater than the number of customers-buyers. The chain on Panel B was probably initiated by the customer-buyer, who was the first to call the dealer and ask to buy securities, though in our data we do not observe the sequence of customer requests before trade execution.

INSERT FIGURE 7 ABOUT HERE

We call a chain complete when we are able to track the movement of a block of instruments between the customer-seller and the customer-buyer. Both intermediation chains shown in Figure 7 are complete, as every dollar of the original customer-sell trade reaches a customer-buyer. Some chains are incomplete because they involve a dealer who holds the volume in inventory or because they occur near the end of our sample period. For each intermediation chain we know how many rounds of intermediation occurred between the two customer trades, the time elapsed between each round, trade volumes and prices at each round, and whether any volume splitting occurred from one round to another. The chain on Panel A has a volume split at the first step and no volume split at the second step; the chain on Panel B has a volume split at the first step. Like the chain on Panel B, many intermediation chains in our sample have no interdealer trades because the same dealer buys from a customer and sells to another customer.

Table 3 reports summary statistics for intermediation chains in our sample. We match 83.7% of the total client-sell volume in Registered instruments and 71% in Rule 144A instruments into complete chains. Intermediation chains are shorter in 144A instruments than in Registered instruments—85% of chains in 144A instruments are one-round chains, with the average chain length conditional on more than one round is only slightly smaller in 144A instruments. In Registered instruments there are longer chains with a larger number of clients. In the Viswanathan and Wang (2004) model, which features uninformed dealers, shorter chains are consistent with greater adverse selection. In the Glode and Opp (2016) model, which features informed dealers, shorter chains are consistent with lower adverse selection. In section 6 we report evidence of more adverse selection in Registered instruments than in Rule 144A instruments, although modest in both markets.

INSERT TABLE 3 ABOUT HERE

Table 4 describes the activity of core and peripheral dealers in intermediation chains. We use eigenvector centrality to classify dealers. The average holding period, calculated as the average difference in execution times for a buy trade and a sell trade executed by the same dealer, is substantially larger for core dealers—core dealers are willing to take on more inventory risk than peripheral dealers. Core dealers in the interdealer market tend to hold inventories longer and are less likely to prearrange trades with their counterparties.

INSERT TABLE 4 ABOUT HERE

We compare average spreads for core dealers with those for peripheral dealers. Table 5 reports the median dealer spreads as well as 75th and 25th percentile customer spreads for smaller-size and larger-size matches. Core dealers provide cheaper intermediation services to their counterparties. In all markets, customer bid-ask spreads are higher for peripheral dealers, with these differences more pronounced for 144A instruments than for Registered instruments.⁷

INSERT TABLE 5 ABOUT HERE

5. Empirical Analysis of Intermediation Chains and Bid-Ask Spreads

We use regressions to study the relation between intermediation chain lengths, time to complete the chain, bid-ask spreads and interdealer network characteristics. Table 6 provides the definitions of the right-hand-side variables we use in the regressions. 32% of intermediation chains in Registered instruments and 42% of intermediation chains in 144A instruments occur within a relatively short time interval, taking less than 15 minutes, while 37% and 31%, respectively take longer than a day. We discard those intermediation chains that take longer than 15 days to complete. Across all instruments the average execution time conditional on the order lasting longer than a day is 6.4

⁷ The 10th percentile of the spread distribution is zero or negative for most types of instruments, indicating that the dealers sometimes make losses on their transactions.

days. We control for possible valuation changes between the execution times by including contemporaneous changes in several market-wide control variables in our regressions: The Baa credit spread, the Treasury term structure slope, the TED spread, and the aggregate mortgage rate obtained from the St. Louis Federal Reserve Economic Data.

INSERT TABLE 6 ABOUT HERE

We allow the regression slope coefficients to be different for Registered and Rule 144A placement types. We include fixed effects for each of the six different collateral types of ABS instruments, for CDO, CBO, and CLO instruments, CMBS interest-only or principal-only (IO/PO) and all other CMBS instruments (P/I), and six different types of CMO tranches separately, which we refer to as subcategories. We combine CBO and CLO into a single subcategory. We cluster standard errors within trade settlement dates, instrument subcategory and placement type.⁸ We check the robustness of our findings using a sample of most traded instruments. The most traded instruments are the top 50 Registered instruments and the top 25 Rule 144A instruments in terms of the number of customer spread observations, excluding instruments with less than 10% of their spreads smaller-size or less than 10% of their spreads larger-size.

Table 7 reports the results from the regressions for the time to complete the intermediation chain in the columns labelled with (1) and the regressions for the chain lengths in the columns labelled with (2). We report the point estimates of the coefficients with standard errors in parentheses.

INSERT TABLE 7 ABOUT HERE

The point estimates of the coefficients on the two vintage dummies is negative in the regressions for the time to complete the intermediation chain in column (1): We observe faster intermediation

⁸ We also experiment by including fixed effects for individual instruments and considering only bid-ask spreads for matches occurring on the same day. Our main results are robust to using individual instrument fixed effects or same-day matches.

chains in higher vintage instruments. These point estimates are of mixed signs and magnitudes in the chain length regressions in column (2): Higher vintage has a mixed effect on how many dealers participate in the chain. The point estimates of the coefficients on active dealers in CUSIP is positive in the full sample in both regressions: Instruments with a greater number of active dealers tend to have longer intermediation chains that take more time to complete. However, we do not see this effect in the most traded instruments.

The regressions in Table 7 document how the expected time to complete the chain and the expected chain length depend on dealer centrality. Across all instruments the coefficient on First in Chain Dealer Centrality is positive in the time to complete regressions and negative in the chain length regressions. When a core dealer buys from a customer, the intermediation chain tends to take a longer time—approximately 14-18% of the standard deviation of the time to complete the chain, or a half a day, even though fewer dealers are involved—a reduction of approximately 30% of the standard deviation of chain length. The regression results suggest that core dealers provide more intermediation services to customers and are more efficient at finding counterparties to complete the chain.

Table 8 reports the results from the regressions for the total customer spreads. The dependent variable is the bid-ask spread, with one observation per pair of matched trades in the sample. We do not report estimates of the bond control variables from Table 6—we find that, generally, older instruments and instruments with riskier cash flows have higher spreads.

INSERT TABLE 8 ABOUT HERE

The point estimate of the coefficient on the number of trades in CUSIP is positive and statistically significant for Registered instruments, but economically small—a one standard deviation increase in the number of trades results in a 9 basis point increase in the bid-ask spread. The effect of the

number of trades is weak and not statistically significant in Rule 144A instruments, in which there is substantially less overall trading activity.

The point estimate of the coefficient on the number of active dealers in CUSIP is negative and statistically significant for Rule 144A instruments in the full sample, and is not statistically significant for Registered instruments and for the sample of most traded instruments. The presence of active dealers in CUSIP enhances liquidity of the not-frequently-traded Rule 144A instruments only, suggesting that dealer competition differs across the two classes of instruments. In Table 1 the average number of active dealers per CUSIP is 4.2 for Registered instruments and 2.4 for Rule 144A instruments, which may explain why the nature of dealer competition differs for Registered and Rule 144A instruments.

In the customer bid-ask spreads regression we include the unexpected time to complete the intermediation chain and the unexpected chain length, calculated as the residuals from the regressions for the time to complete the intermediation chain and the regressions for the chain lengths in Table 7. Dealer centrality is correlated with chain length and expected time to complete the chain. When a customer chooses between a core or a peripheral dealer, she may anticipate that the intermediation chain that follows will depend on her choice. To measure the total effect of dealer centrality on the spreads we include the part of chain length and time to complete the chain unrelated to dealer centrality in the customer spread regressions.

The point estimates of the coefficients on the unexpected chain length and the unexpected time to complete the chain are positive and statistically significant. A one standard deviation increase in the unexpected chain length increases the transaction costs by 40 to 100 basis points and the effect of a one standard deviation increase in the unexpected time to complete the chain is between 9 and 27 basis points depending on the instrument type in the full sample.

The regression documents volume discounts. The point estimates of the coefficients on the smaller-size trade dummy is positive and statistically significant for all types of instruments in the full sample. The difference in spreads between smaller-size trades and larger-size trades reported in Table 2 is striking, and remains robust when we control for various bond and trade characteristics. The regression documents 50 to 78 basis points difference between smaller-size spreads and larger-size spreads. The point estimate of the coefficient on the average trade volume per CUSIP is negative and statistically significant for Registered instruments, indicating that instruments with larger trades tend to have small spreads. The coefficient on the deviation of trade volume from average is negative and statistically significant for Registered instruments, indicating that when the matched trade is larger than average for that instrument, the match will have a lower spread reflecting a volume discount. The volume discount is weaker for Rule 144A instruments because this is entirely an institutional market.

In typical equity markets, larger trades tend to have larger spreads, with the standard explanation being that larger trades carry more information so that dealers face higher adverse selection costs on larger trades. In many bond markets, smaller trades have larger spreads, with the standard explanation being that smaller trades tend to proxy for less sophisticated customers so that dealers have greater bargaining power in smaller trades and so are able to earn higher spreads on smaller trades. The securitized markets we analyze resemble typical bond markets with respect to the effects of volume on spreads.

The point estimate on the first dealer in chain centrality dummy is negative and statistically significant for all instruments. The average spread is 40 to 64 basis points lower if the client contacts a dealer who is more central in the interdealer network. The point estimate on the core dealer later in chain is negative and statistically significant for all instruments. The average spread

is approximately 20 basis points lower for Registered instruments and more than 100 basis points lower for Rule 144A instruments if the inventory passes through a dealer who is more central in the interdealer network. As reported in Table 7 the coefficient on the first dealer in chain centrality dummy is negative in the chain length regressions, meaning that core dealers tend to shorten the intermediation chains.

When we rerun the regressions reported in Table 8 using total chain length and total time to complete rather than the unexpected portions, the coefficients on the dealer centrality dummy is still negative, but smaller in magnitude.⁹ Core dealers tend to have lower spreads partially through their effect on expected chain length indicating that core dealers perform a valuable function by enhancing the linkages in the network and shortening intermediation chains. Our theoretical model where core and peripheral dealers trade with different customer clienteles predicts that having relatively sophisticated customers in the securitization markets can explain why spreads are relatively lower with core dealers than with peripheral dealers.

Our regression results are robust to including the interaction terms between smaller-size trade dummy, deviation of trade volume, dealer centrality, and the interaction terms between smaller-size trade dummy and the unexpected chain length and the unexpected time to complete the chain. We find that the volume discount is weaker for core dealers than for peripheral dealers. The finding is robust when we look at the most traded Registered instruments. The results are also robust to including the same-day matches only in the customer spread regression and also to including security fixed-effects.¹⁰

Several dealers may contribute their share to the total bid-ask spread and the intermediation cost for customers. When a customer contacts a dealer to sell securities, the exact path of the securities

⁹ These results are available in an Online Appendix.

¹⁰ These results are available in an Online Appendix.

within the interdealer network and the length of the intermediation chain are uncertain. We use intermediation chains with at least two dealers to study how interdealer trades contribute to the costs of the intermediation service. We repeat the regression analysis with additional controls for the characteristics of the intermediation chain.

Table 9 reports the results from the regressions for the interdealer bid-ask spreads for intermediation chains that include at least two dealers. We include the same controls for instrument characteristics as in Table 8 but do not report them in Table 9 since the findings are generally the same for these control variables.

INSERT TABLE 9 ABOUT HERE

The point estimate on the sell to dummy is positive and statistically significant for Registered instruments. Dealers earn 16 basis points higher spreads when they buy from a dealer and sell to a customer, than when they buy from a dealer and sell to a dealer. The point estimate on the sell to dummy is negative in Rule 144A instruments. In Rule 144A markets dealers earn the highest markup when they buy from a dealer and sell to a dealer—12 to 14 basis points higher than when selling to a customer. The point estimate on the buy from dummy is negative for all instruments. Dealers earn 27 to 58 basis points lower spreads when they buy from a customer and sell to a dealer, than when they buy from a dealer and sell to a dealer. These findings suggest being the first dealer in a long chain is as profitable for the dealer as being a dealer in the middle of the chain. The point estimate on the split dummy is positive for all instruments and statistically significant for Registered instruments. Dealers earn a 24 basis point premium in Registered instruments when they split volume or arrange a reverse-request transaction.

The point estimate on the dealer centrality dummy is negative and statistically significant for Rule 144A instruments in the full sample and of mixed sign and weak statistical significance for other

instruments and for the most traded sample. Thus, except in Rule 144A instruments, we do not observe a centrality discount on the interdealer market. Core dealers do not offer discounts to other dealers on the interdealer market. The point estimate on the chain length is negative and statistically significant—as an intermediation chain gets longer, the spread of each participating dealer gets lower. This contrasts the effect of the unexpected chain length on the total customer spread in Table 7. The point estimate on the time to complete chain is positive and statistically significant for all instruments in the full sample, and of mixed sign and weak statistical significance for the most traded sample. When the holding period of a dealer increases the spread tends to increase, especially in the infrequently traded instruments.

The interdealer spreads regression confirms the volume discounts in the interdealer market. The point estimate on the smaller-size trade dummy is positive and statistically significant in the full sample, the point estimates on the average trade volume per CUSIP and the deviation of trade volume are negative and statistically significant in the full sample.

We use quantile regressions of customer bid-ask spreads on dealer centrality to study spread dispersions. Table 10 reports unconditional means, standard deviations of the distribution of customer bid-ask spreads as well as the number of spread observations for the core dealers and peripheral dealers. The standard deviation of spreads for core dealers is lower than for peripheral dealers. The quantile regressions that control for instrument characteristics and market-wide control variables to check the robustness of relative dispersions of bid-ask spreads between core and peripheral dealers. In the bottom of Table 10 we report the coefficients on dealer centrality in a quantile regression of the spreads where we control for instrument characteristics and the market-wide control variables described in Section 5. The coefficients on dealer centrality are negative with the largest negative effect observed for the 90th percentile of bid-ask spreads and the weakest

negative effect for the 10th percentile of bid-ask spreads. Controlling instrument characteristics and market-wide movements, the distribution of bid-ask spreads is less dispersed at the center of the dealer network than at the periphery.

INSERT TABLE 10 ABOUT HERE

To summarize, we document a centrality discount—both total client bid-ask spreads and dealer spreads tend to be lower when a core dealer intermediates trades. The centrality discount in the market for securitizations contrasts with the centrality premium reported by Li and Schürhoff (2016) for municipal bond markets. The model we develop in Section 2 explains why spreads are relatively lower with core dealers than with peripheral dealers in the securitization markets, while spreads are relatively higher with core dealers than with peripheral dealers in municipal bond markets. The model also predicts a lower bid-ask spread dispersion when there is enough dispersion in the dealers’ trade execution speeds. We document a lower bid-ask spread dispersion for core dealers.

6. Price Discovery in the Network

Our empirical finding that core dealers charge lower bid-ask spreads on average than peripheral dealers is also consistent with the model in Babus and Kondor (2016), in which core dealers charge lower bid-ask spreads due to lower exposure to adverse selection risk. We use a panel VAR model to document the nature of the price discovery process in the securitization trading networks.

We estimate a panel VAR model with fixed effects and 5 lags, based on Hasbrouck (2007) and the GMM estimator of Abrigo and Love (2015) using customer trades in each instrument i . Let t be the event time when a trade occurs, and let r_{it} be the log-return between trades at t and $t - 1$ in

instrument i . Define q_{it} as the signed logarithm of trade volume and c_{it} the signed dealer eigenvector centrality discussed in Section 4. The panel VAR specification is:

$$\begin{aligned}
r_{it} &= \sum_{j=1}^5 \alpha_{r,j} r_{i,t-j} + \sum_{j=1}^5 \beta_{r,j} q_{i,t-j} + \sum_{j=1}^5 \gamma_{r,j} c_{i,t-j} + u_{r,i} + \varepsilon_{r,it}, \\
q_{it} &= \sum_{j=1}^5 \alpha_{q,j} r_{i,t-j} + \sum_{j=1}^5 \beta_{q,j} q_{i,t-j} + \sum_{j=1}^5 \gamma_{q,j} c_{i,t-j} + u_{q,i} + \varepsilon_{q,it}, \\
c_{it} &= \sum_{j=1}^5 \alpha_{c,j} r_{i,t-j} + \sum_{j=1}^5 \beta_{c,j} q_{i,t-j} + \sum_{j=1}^5 \gamma_{c,j} c_{i,t-j} + u_{c,i} + \varepsilon_{c,it}.
\end{aligned} \tag{7}$$

We sort instruments in terms of the number of customer spread observations, excluding instruments with less than 10% of their spreads smaller-size or less than 10% of their spreads larger-size. We pick the most traded instruments as the top 50 Registered instruments and the top 25 Rule 144A instruments in that sorting, the next 50 Registered instruments and 25 Rule 144A instruments are the second-most traded instruments. We estimate equation (7) for the two resulting samples of instruments. We use the coefficient estimates to compute the informational attributions of order flow and price changes as in Hasbrouck (2007, chapter 9) to proxy for information content. Abrigo and Love (2015) provide an algorithm to do this in a panel VAR setting. Price changes are ordered last in the VAR ordering and we consider alternative orderings of the remaining variables.

Panel A of Table 11 reports our estimation results when the ordering of the variables is: signed log-volume, dealer centrality dummy, log-return. In the most-traded Registered instruments 12.1% of the long-term variance in price changes is explained by signed log-volume, and 0.7% is explained by the signed dealer centrality dummy. In the most-traded Rule 144A instruments 2.8% of the long-term variance in price changes is explained by signed log-volume, and 2.4% is explained by the signed dealer centrality dummy. The effect of signed dealer centrality dummy is economically small and not significant for both types of instruments, while the effect of signed log-volume is significant in Registered instruments. We conclude that there is relatively stronger evidence of adverse

selection in Registered instruments compared to Rule 144A instruments—the order flow has a stronger long-term impact on price changes, however there is no evidence that dealer centrality contains significant additional information about future price changes in both types of instruments. The findings are robust for the second-most traded instruments.¹¹

INSERT TABLE 11 ABOUT HERE

We also report results using an alternative sorting of variables in the panel VAR. Panel B of Table 11 reports our estimation results when the ordering of the innovations is: dealer centrality dummy, signed log-volume, log-return. Our results for Rule 144A instruments are robust, while in Registered instruments we observe a stronger effect of dealer centrality dummy—it explains 6.7% of the long-term variance in price changes for the most traded instruments, and 10.2% of the long-term variance in price changes for the second-most traded instruments. The incremental explanatory power of signed log-volume is non-trivial, 6.1% and 5.0% for most traded and second-most traded Registered instruments, respectively.

From Table 3, Registered instruments have longer chains with a larger number of clients than Rule 144A instruments. From the VAR analysis we find evidence of more adverse selection in Registered instruments. The finding is consistent with the Glode and Opp (2016) model in which longer chains are consistent with more adverse selection.

7. Concluding Comments

We utilize data on customer and dealer trades in securitization markets to study the nature of dealer networks, intermediation chains, and how bid-ask spreads vary within the trading network. Interdealer networks have a core-peripheral structure in the market for securitizations, similar to the

¹¹ The third-most traded instruments have very few observations in the Rule 144A sample, as there is less than 20 customer trades per instrument.

municipal bond markets (Li and Schürhoff 2016). We document a negative relationship between interconnectedness of dealers and their bid-ask spreads.

Our evidence contrasts with the empirical findings in municipal bond markets, where a positive relationship arises between dealers' centrality and bid-ask spreads and prices are more efficient when the trades are intermediated by core dealers, as documented by Li and Schürhoff (2016). We present a model where enough customer and dealer heterogeneity can result in a centrality discount. The difference across markets may reflect the presence of more sophisticated traders in the securitization markets than in the municipal bond markets.

Theoretical work studying over-the-counter markets predicts that customers that trade with more interconnected dealers with higher trade execution efficiency face lower bid-ask spreads on average in equilibrium, as in Babus and Kondor (2016), Malamud and Rostek (2016), Neklyudov (2013). We estimate a panel VAR model, finding that dealers' centrality does not influence the price discovery process in the securitization markets.

Our empirical findings inform the theory of over-the-counter markets and offer a basis for exploring different theoretical models. It is important to understand the microeconomic aspects of the trading process in the presence of the dramatic disclosure differences between Registered and 144A instruments. There is less adverse selection in 144A instruments, even though 144A instruments have less disclosure requirements. The structure of our data allows us to identify different counterparties and construct trading networks, offering a natural environment to perform network analysis. Network analysis can enhance our understanding of intermediation patterns for dealer markets and concentrations of risk more broadly, including systemic risks.

References

- Abrigo, Michael, and Inessa Love, 2015, “Estimation of Panel Vector Autoregression in Stata: A Package of Programs,” *working paper*, University of Hawaii.
- Adamic, Lada, Celso Brunetti, Jeffrey H. Harris, and Andrei Kirilenko, 2010, “Trading Networks,” *working paper*, American University.
- Asquith, Paul, Thom Covert, and Parag Pathak, 2013, “The Effects of Mandatory Transparency in Financial Market Design: Evidence from the Corporate Bond Market,” *working paper*, National Bureau of Economic Research.
- Atanasov, Vladimir, and John J. Merrick, Jr., 2012, “Liquidity and Value in the Deep vs. Shallow Ends of Mortgage-Backed Securities Pools,” *working paper*, College of William and Mary.
- Atkeson, Andrew, Andrea Eisfeldt, and Pierre-Olivier Weill, 2015, “Entry and Exit in OTC Derivatives Markets,” *Econometrica* 83.6: 2231-2292.
- Babus, Ana, and Peter Kondor, 2016, “Trading and Information Diffusion in OTC Markets,” *working paper*, Federal Reserve Bank of Chicago and Central European University.
- Bernhardt, Dan, Vladimir Dvoracek, Eric Hughson, and Ingrid M Werner, 2005, “Why do larger orders receive discounts on the London Stock Exchange?” *Review of Financial Studies* 18: 1343–1368.
- Bessembinder, Hendrik, and William Maxwell, 2008, “Markets: Transparency and the Corporate Bond Market,” *Journal of Economic Perspectives* 22: 217-234.
- Bessembinder, Hendrik, William Maxwell, and Kumar Venkataraman, 2006, “Market Transparency, Liquidity Externalities, and Institutional Trading Costs in Corporate Bonds,” *Journal of Financial Economics* 82: 251-288.
- Bessembinder, Hendrik, William Maxwell, and Kumar Venkataraman, 2013, “Trading Activity and Transactions Costs in Structured Credit Products,” *Financial Analysts Journal* 69.6: 55-67.
- Chang, Briana, and Shengxing Zhang, 2016, “Endogenous market making and network formation,” *working paper*, University of Wisconsin-Madison and London School of Economics.
- Di Maggio, Marco, Amir Kermani, and Zhaogang Song, 2016, “The value of trading relationships in turbulent times,” *Journal of Financial Economics*, forthcoming.
- Duffie, Darrell, Nicolae Gârleanu, and Lasse Heje Pedersen, 2005, “Over-the-Counter Markets,” *Econometrica* 73: 1815–1847.

Duffie, Darrell, Nicolae Gârleanu, and Lasse Heje Pedersen, 2007, “Valuation in over-the-counter markets,” *Review of Financial Studies* 20: 1865–1900.

Edwards, Amy K., Lawrence E. Harris, and Michael S. Piwowar, 2007, “Corporate Bond Market Transaction Costs and Transparency,” *Journal of Finance* 62: 1421-1451.

Financial Crisis Inquiry Commission, 2011, “The financial crisis inquiry report: Final report of the national commission on the causes of the financial and economic crisis in the United States,” *Public Affairs*.

Friewald, Niles, Rainer Jankowitsch and Marti G. Subrahmanyam, 2016, “Transparency and Liquidity in the Structured Product Market,” *working paper*, New York University.

Glode, Vincent, and Christian Opp, 2016, “Asymmetric Information and Intermediation Chains,” *American Economic Review* 106: 2699-2721.

Goldstein, Michael, Edith Hotchkiss, and Erik Sirri, 2007, “Transparency and Liquidity: A Controlled Experiment on Corporate Bonds,” *Review of Financial Studies* 20: 235–273.

Green, Richard C., Burton Hollifield and Norman Schürhoff, 2007, “Financial Intermediation and the Costs of Trading in an Opaque Market,” *Review of Financial Studies* 20: 275-314.

Harris, Lawrence E., and Michael S. Piwowar, 2006, “Secondary Trading Costs in the Municipal Bond Market,” *Journal of Finance* 61: 1361-1397.

Hasbrouck, Joel, 2007, “*Empirical Market Microstructure: The Institutions, Economics, and Econometrics of Securities Trading*,” Oxford University Press.

Hugonnier, Julien, Benjamin Lester, and Pierre-Olivier Weill, 2015, “Heterogeneity in Decentralized Asset Markets,” *working paper*, Federal Reserve Bank of Philadelphia.

Jackson, Matthew, 2008, “*Social and economic networks*,” Princeton University Press.

Lester, Benjamin, Guillaume Rocheteau, and Pierre-Olivier Weill, 2015, “Competing for order flow in OTC markets,” *Journal of Money, Credit and Banking* 47: 77-126.

Li, Dan, and Norman Schürhoff, 2016, “Dealer Networks,” *working paper*, Federal Reserve Board and University of Lausanne.

Malamud, Semyon, and Marzena J. Rostek, 2016, “Decentralized Exchange,” *working paper*, EPFL Lausanne and University of Wisconsin-Madison.

Neklyudov, Artem, 2013, “Bid-Ask Spreads and the Over-the-Counter Interdealer Markets: Core and Peripheral Dealers,” *working paper*, University of Lausanne.

Neklyudov, Artem, and Batchimeg Sambalaibat, 2016, “Endogenous Specialization and Dealer

Networks,” *working paper*, University of Lausanne and Indiana University.

Segoviano, Miguel, Jones Bradley, Peter Lindner, and Johannes Blankenheim, 2015, “Securitization: The Road Ahead,” *IMF Staff Discussion Note*.

Shen, Ji, Bin Wei, and Hongjun Yan, 2016, “Financial Intermediation Chains in an OTC Market,” *working paper*, DePaul University.

SIFMA, 2016, “Statistics,” Accessed December 27. <http://www.sifma.org/research/statistics.aspx>.

Üslü, Semih, 2015, “Pricing and liquidity in decentralized asset markets,” *working paper*, University of California Los Angeles.

Viswanathan, S., and James JD Wang, 2004, “Inter-Dealer Trading in Financial Markets,” *Journal of Business* 77: 987-1040.

Weller, Brian M., 2013, “Fast and Slow Liquidity,” *working paper*, Northwestern University.

Zhong, Zhuo, 2014, “The risk sharing benefit versus the collateral cost: The formation of the inter-dealer network in over-the-counter trading,” *working paper*, The University of Melbourne.

APPENDIX A: Proof of Propositions 1 and 2

Case 1: Consider the case when $U \leq u_p$. In this case all buyers are indifferent between core and peripheral dealers. The transaction price with peripheral dealers is uniformly distributed on the interval

$$\left[\frac{1}{r} - \left(1 + r/\lambda_p\right) U, \frac{1}{r} \right], \quad (\text{B1})$$

and the transaction price with core dealers is uniformly distributed on the interval

$$\left[\frac{1}{r} - \left(1 + r/\lambda_c\right) U, \frac{1}{r} \right]. \quad (\text{B2})$$

Since $\lambda_c > \lambda_p$, average ask prices are higher on average and price dispersion is lower for trades with core dealers than for trades with peripheral dealers.

Case 2: Consider the case when $u_p < U \leq u_c$. In this case the buyers with reservation utility $u_p < u \leq U$ do not trade with the peripheral dealers, but always trade with the core dealers. The buyers with reservation utility $u \leq u_p$ trade with the peripheral dealers with probability π and trade with the core dealers with probability $1 - \pi$. The transaction price with peripheral dealers is uniformly distributed on the interval

$$\left[L, \frac{1}{r} \right], \quad (\text{B3})$$

and the transaction price with core dealers is a mixture of two uniform distributions with the weight

$\frac{U - u_p}{(1 - \pi)u_p + U - u_p}$ on a uniform distribution on the interval

$$\left[\frac{1}{r} - \left(1 + r/\lambda_c\right) U, \frac{1}{r} - \left(1 + r/\lambda_c\right) u_p \right], \quad (\text{B4})$$

and the weight $\frac{(1 - \pi)u_p}{(1 - \pi)u_p + U - u_p}$ on a uniform distribution on the interval

$$\left[\frac{1}{r} - \left(1 + r/\lambda_c\right) u_p, \frac{1}{r} \right]. \quad (\text{B5})$$

Since $U \leq u_c$ we have $\frac{1}{r} - \left(1 + r/\lambda_c\right)U \geq L$, so transaction prices have a wider range of possible values among the peripheral dealers than among the core dealers. The average prices are higher on average among peripheral dealers when

$$\frac{1}{2} \left(\frac{1}{r} - \left(1 + r/\lambda_c\right)u_p \right) - \frac{U - u_p}{(1-\pi)u_p + U - u_p} \left(\frac{1}{2} \left(1 + r/\lambda_c\right)U \right) \leq \frac{1}{2}L. \quad (\text{B6})$$

When $U = u_c$ inequality (B6) always holds. When $U = u_p$ inequality (B6) never holds. The left-hand side of the inequality is monotonically decreasing in U and decreasing in π , so we conclude that average ask prices can be higher or lower for trades with core dealers than for trades with peripheral dealers. Moreover, there exists a threshold value of probability $\pi^*(U)$, which is decreasing in U , such that for all $\pi < \pi^*(U)$ the inequality (B6) does not hold.

Case 3: Consider the case when $u_p < u_c < U$. The transactions prices with peripheral dealers are uniformly distributed on the interval

$$\left[L, \frac{1}{r} \right], \quad (\text{B7})$$

and the transactions prices with core dealers are a mixture of two distributions with the weight $\frac{u_c - u_p}{(1-\pi)u_p + u_c - u_p}$ on a uniform distribution on the interval

$$\left[L, \frac{1}{r} - \left(1 + r/\lambda_c\right)u_p \right], \quad (\text{B8})$$

and the weight $\frac{(1-\pi)u_p}{(1-\pi)u_p + u_c - u_p}$ on a uniform distribution on the interval

$$\left[\frac{1}{r} - \left(1 + r/\lambda_c\right)u_p, \frac{1}{r} \right]. \quad (\text{B9})$$

The average prices are higher on average among peripheral dealers when

$$\frac{1}{2} \left(\frac{1}{r} - \left(1 + r/\lambda_c\right)u_p \right) - \frac{u_c - u_p}{(1-\pi)u_p + u_c - u_p} \left(\frac{1}{2} \left(\frac{1}{r} - L \right) \right) \leq \frac{1}{2}L. \quad (\text{B10})$$

The left-hand side of inequality (B10) is monotonically increasing in π , and for $\pi = 0$ (B10) holds as an equality. Thus for any $0 < \pi < 1$ we have a strict inequality and the average ask prices are lower for trades with core dealers than for trades with peripheral dealers.

We use the distribution for the transaction price with core dealers in equations (B8) and (B9) to compute the price dispersion as a function of π . The price dispersion for trades with core dealers is a quadratic function of π , which is maximized when:

$$\pi^* = \frac{\lambda_c(r+\lambda_p)(r(2\lambda_p-\lambda_c)+\lambda_c\lambda_p)}{\lambda_p(r+\lambda_c)(r(2\lambda_c-\lambda_p)+\lambda_c\lambda_p)}. \quad (\text{B11})$$

When $\pi \rightarrow 0$ then the price dispersions for trades with core dealers than for trades with peripheral dealers are equal. When

$$\lambda_c > 2\lambda_p \text{ and } r \geq \lambda_c\lambda_p/(\lambda_c - 2\lambda_p), \quad (\text{B12})$$

then $\pi^* \leq 0$, and the derivative of the price dispersion for trades with core dealers with respect to π is negative. These two inequalities are equivalent to:

$$u_p \leq u_c/2. \quad (\text{B13})$$

Therefore, the price dispersion is lower for trades with core dealers than for trades with peripheral dealers for any values of π . When the inequality (B13) does not hold, the derivative of the price dispersion for trades with core dealers with respect to π is positive at $\pi = 0$, so the price dispersion will be higher for trades with core dealers than for trades with peripheral dealers when π is low enough. Since the price dispersion for trades with core dealers is the highest in Case 3, (B13) is the sufficient condition for all cases.

QED.

APPENDIX B: Data Construction

Our sample contains trading activity data ranging from May 16, 2011 to February 29, 2012 in several categories of securitized products: ABS, CDO, CMBS, CMO, MBS and TBA, and security characteristics for all securities subject to the FINRA reporting requirements. We limit our attention to ABS, CMBS, CDO and Non-Agency CMO securitizations because these categories have 144A instruments in our sample.

Registered securities have a registration statement filed with the Securities and Exchange Commission (SEC), which includes a prospectus and several registration forms covering various aspects of the offering, can be traded publicly. However, some securities can trade without being registered. One such example is Rule 144A adopted by the SEC to allow trades of unregistered securities between relatively large institutions. Rule 144A permits sales to qualified institutional buyers (QIBs) and is often used by issuers to offer QIBs debt securities and to avoid the requirement of public information disclosure. The documentation that follows the offering of Rule 144A securities includes the prospectus and the issuer's financial statements available upon request by the security holder. Holders of Rule 144A securities are not entitled to the same level of legal protection and due diligence defense as holders of registered securities as there is reduced potential for liability for the issuer.

Rule 144A securities are considered restricted securities for a period between six months and one year after their initial offering, depending on whether the issuer is a reporting company or not. The securities can become unrestricted after the holding period elapses, can be traded outside of the QIBs market, and in this case will be assigned a different unrestricted CUSIP identifier. In our sample we track a security that changes status from Rule 144A to Registered by matching a Rule 144A CUSIP to a corresponding Registered CUSIP using the following five criteria: same maturity,

coupon rate, issuer name, coupon and interest type. Out of 38,768 Rule 144A instruments we find 48 ABS instruments, 2 CMBS instruments, and 4 CMO instruments that dropped restricted status and these instruments have few trades in our sample.

Our trading activity data contains trades in both agency and non-agency CMO instruments. We use the “ISSR_SHORT_NM” variable in the security characteristics dataset to classify all CMO instruments issued by Freddie Mac, Fannie Mae, or Ginnie Mae as Agency CMO instruments and drop these instruments from our trading activity data. Our initial sample consists of 12,257 ABS instruments, 7,424 CDO instruments, 13,353 CMBS instruments, and 75,240 Non-Agency CMO instruments. Before cleaning there are 30,976 instruments with at least one trade report, and 452,555 trade reports in total.

We perform several rounds of data cleaning to obtain our final sample of trades. We drop instruments that mature before or on February 29, 2012, which is the last day of our sample period. There are 275 such instruments and 1,096 trade reports associated with them. Then we perform cleaning of trades that were cancelled, modified, or interdealer trades that were reported twice. For some trade records traders entered incorrect trade information or canceled previous trades. For those cases we have reports marked as “Corrected Trades”, “Trade Cancels” or “Cancels”, and “Historical Reversals” if the correction was reported on a different trading day. The FINRA reporting rule is that each interdealer trade must be reported by both sides to the trade, leading to double reporting in our sample, with a few exceptions. Customer trades and locked-in trades are reported once, and in our data we observe whether the trade is a customer trade or a locked-in trade. We follow procedures similar to those used by Asquith, Covert, and Pathak (2013) for cleaning such trades in the enhanced TRACE dataset for corporate bonds. As a result of these procedures, there are 30,822 instruments with at least one trade report, and 366,445 trade reports in total. Sometimes a dealer

acts as an agent and executes an agency trade on behalf of another dealer. There are 23,677 such agency trades in our sample. We use both agency and principal trades to recover the interdealer trading networks.¹²

We used the proprietary list of CUSIPs provided by FINRA to obtain Moody's ratings for these instruments on the Moody's corporate website. For those instruments not rated by Moody's in our sample, FINRA provides information whether the instruments are investment grade or high yield. We classify unrated instruments as high yield rather than investment grade instruments. We use the security characteristics data provided by FINRA and construct the time-series of coupon rates and prepayment factors, the type of collateral, maturity, original balance, type of placement (Registered or 144A), and coupon type (fixed or floating) for each instrument.

We use interdealer trades to construct dealer networks, separately for Registered and Rule 144A instruments. In Registered instruments, there are 625 dealers participating in 77,464 interdealer trades, out of 314,763 trades in total. In Rule 144A instruments, there are 332 dealers participating in 9,898 interdealer trades, out of 51,682 trades in total. For each pair of dealers, we calculate the total volume traded over the sample period; we keep those pairs with more than \$5M traded as edges in a dealer network. Thus we keep 2,304 out of 5,232 pairs of dealers in Registered instruments and 1,022 out of 1,659 pairs of dealers in Rule 144A instruments. Our results are robust to an alternative cutoff choice of \$10M rather than \$5M.

We use trades with customers and interdealer trades to construct intermediation trees, separately for each CUSIP. An intermediation tree is a directed graph, in which each node is a customer or a dealer, and each link is a trade. For example, we construct the tree depicted in panel A of Figure 7

¹² We use only principal trades when we construct intermediation trees and report chain lengths.

from three trades: A customer sells \$10M of original balance to dealer A, dealer A sells \$5M to one customer, and sells \$5M to another customer. Our matching algorithm ensures that for each link of the tree there are no alternative links that reduce the total execution time of all trades in the tree. The algorithm imposes the constraint that a dealer sell cannot occur earlier than one day before the corresponding dealer buy.

Our matching algorithm has two rounds. In the first round we take all trades where a customer sells to a dealer and look for corresponding trades where the same dealer sells to a customer or another dealer. When we do not find any matching trade, we stop the algorithm for that initial customer sell trade. When we find several matching trades, we choose the trade with the closest execution timestamp to the preceding trade and perform LIFO matching. When the volume of the trade we chose as a match exceeds the original volume of the preceding trade, we split the trade in two parts with one having the same volume as the preceding trade and the other having the remaining volume. We keep the trade with the remaining volume in the sample of unmatched trades. When the volume of the trade is less than the original volume of the preceding trade, we save it as a partial match with a volume split and keep running the algorithm to match the remaining part of the preceding trade. We stop the first round when no new matches can be produced for any trade.

Sometimes dealers participate in reverse-request transactions, in which there are multiple customer-sell trades, the dealer aggregates the volume of these trades, and there is one subsequent trade. Our LIFO method results in a separate tree for each of the customer-sell trade in a reverse-request transaction, and all such trees have a common ending. In the second round we combine all such trees.

We obtain 19,078 intermediation trees in ABS instruments, 2,721 trees in CDO instruments, 21,680 trees in CMBS instruments, and 54,243 trees in Non-Agency CMO instruments. Figure 7 illustrates two common types of intermediation trees we identify in our data, and Table 3 reports descriptive statistics for the intermediation trees in our sample.

We adjust prices for coupon and factor payments that happened between the settlement time of a trade and the settlement time of a buy from customer (the beginning of the chain), as well as for accrued interest. For each chain we compute two types of bid-ask spread measures: total client bid-ask spread and dealer-specific spread—both measured per \$100 of current value.

We compute bid-ask spread using the two prices P_{ask} and P_{bid} and the information on factor and coupon payments. Consider the case when the settlement date of the ask trade occurs after the settlement date of the bid trade. Let T be the number of calendar days in between the two dates, c be the annual dollar coupon amount per \$100 of original balance, f_{ask} be the factor at the time of the ask trade, and f_{bid} be the factor at the time of the bid trade. The bid-ask spread is computed as:

$$Spread = \frac{100(P_{ask} f_{ask} + adj - P_{bid} f_{bid})}{\frac{1}{2}(P_{ask} f_{ask} + P_{bid} f_{bid} + adj)}, \quad (A1)$$

where $adj = c \frac{T}{360} f_{bid} + prepayment$.

We use the fair-pricing condition:

$$\frac{prepayment}{P_{ask}} = (f_{bid} - f_{ask}). \quad (A2)$$

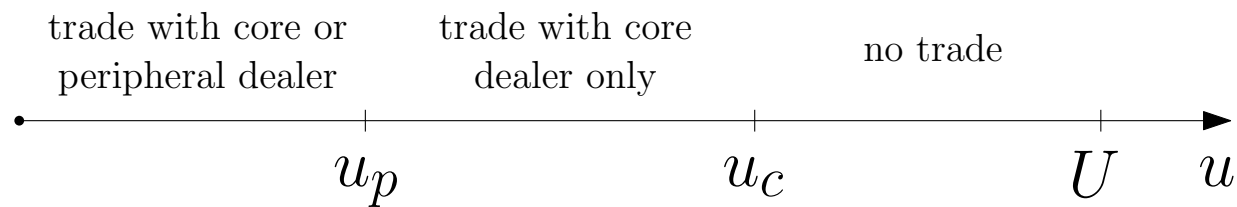
When the fair-pricing condition (A2) holds, the bid-ask spread equation (A1) simplifies to:

$$Spread = \frac{100(P_{ask} - P_{bid} + c \frac{T}{365})}{\frac{1}{2}(P_{ask} + P_{bid} + c \frac{T}{365})}. \quad (A3)$$

Both formulas result in similar spread distributions. The fair-pricing condition is thus a relatively good approximation for those matches that involve factor payments in between the two settlement

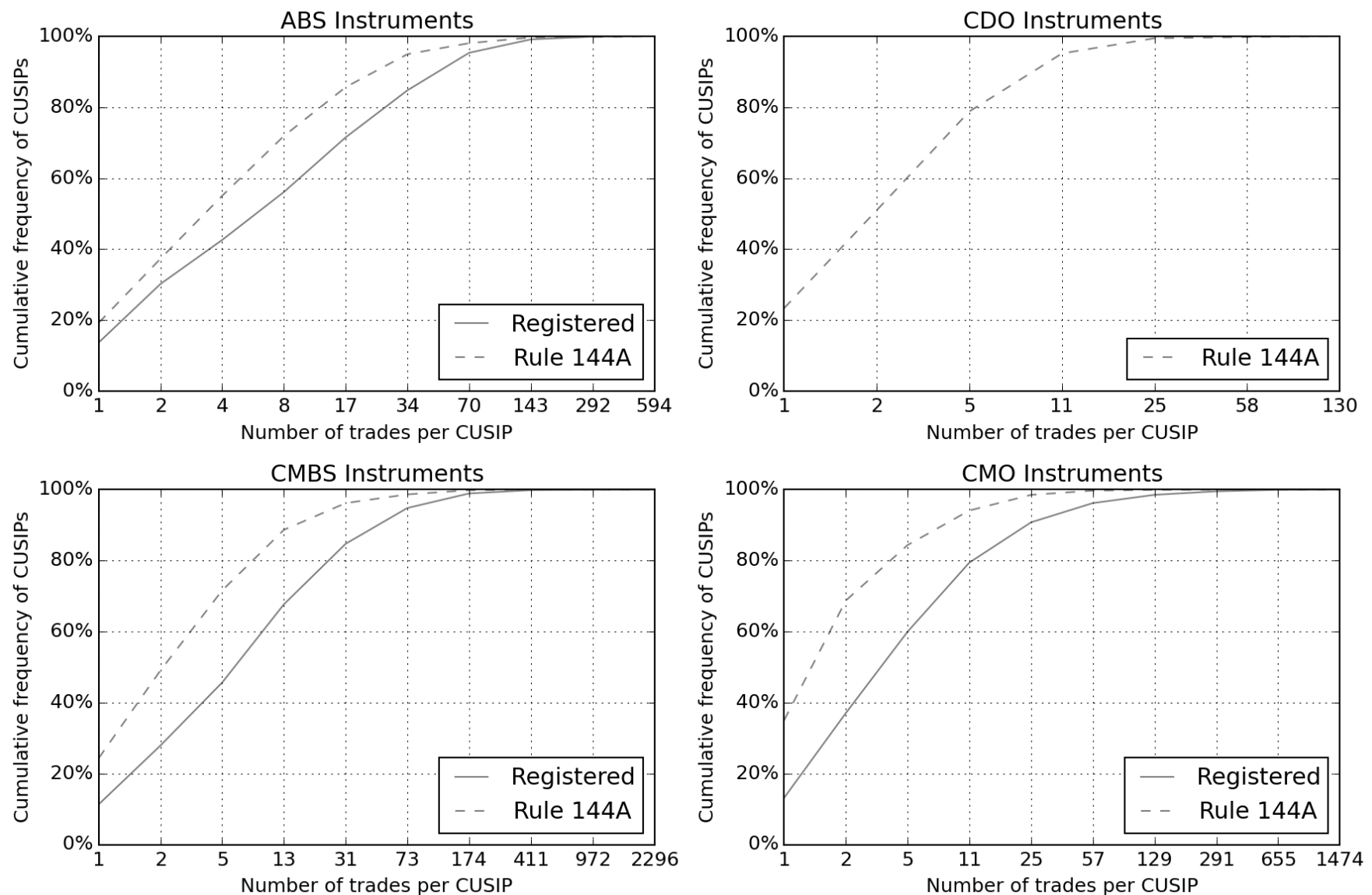
dates. Few spread observations are outliers so we winsorize 1% of the upper and lower tails of the distribution of total client spreads within each collateral type, placement type, and credit rating.

Figure 1: The Equilibrium Decision Rules



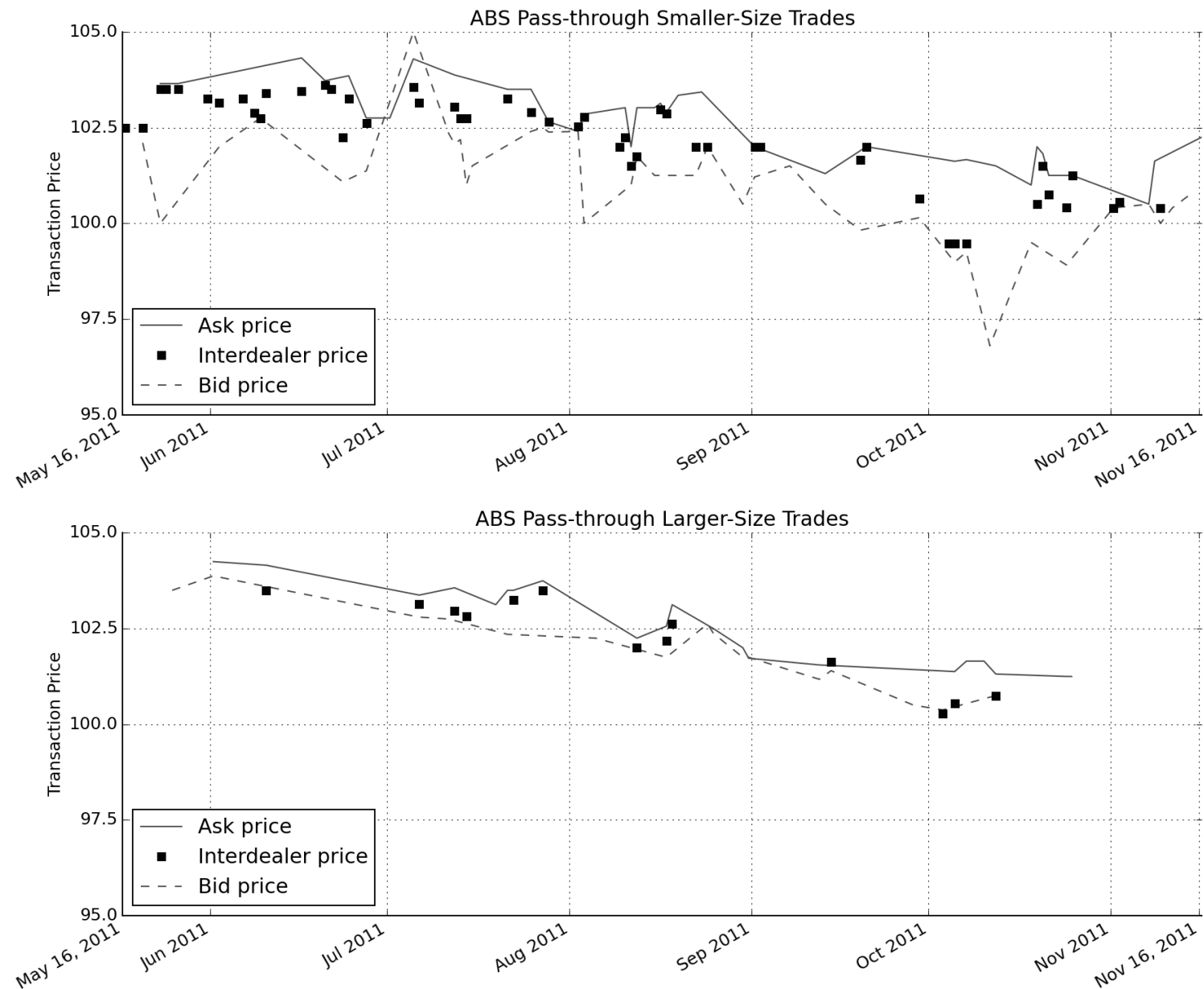
The figure provides the graphical description of the equilibrium decision rules of the buyers. The plot shows the decision rules for buyers as a function of the realized reservation utility u .

Figure 2: Trading Frequencies for Securitizations



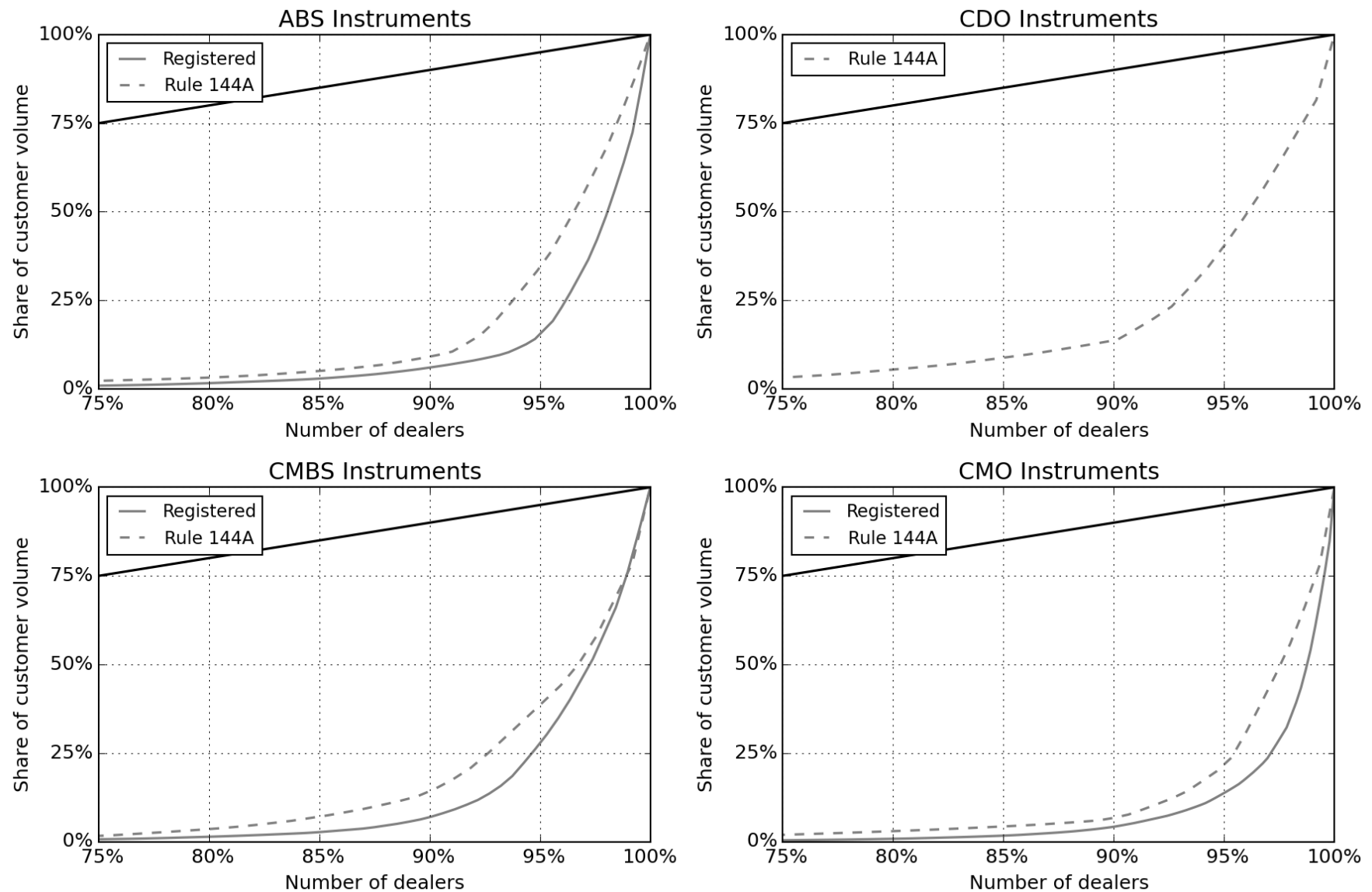
Number of trade records includes both trades with customers and interdealer trades after the trade data cleaning discussed in the Appendix B. The x-axis is labeled by the number of trade reports per CUSIP. There are nine equally spaced and log-scaled intervals for ABS, CMBS and CMO instruments and six intervals for CDO instruments. The y-axis shows the cumulative frequencies of CUSIPs separately for Registered and Rule 144A instruments, except the CDO category which contains Rule 144A instruments only. The sample period is from May 16, 2011 to February 29, 2012 (208 trading days). We show only those instruments with at least one trade in our sample.

Figure 3: An Example of Transaction Prices



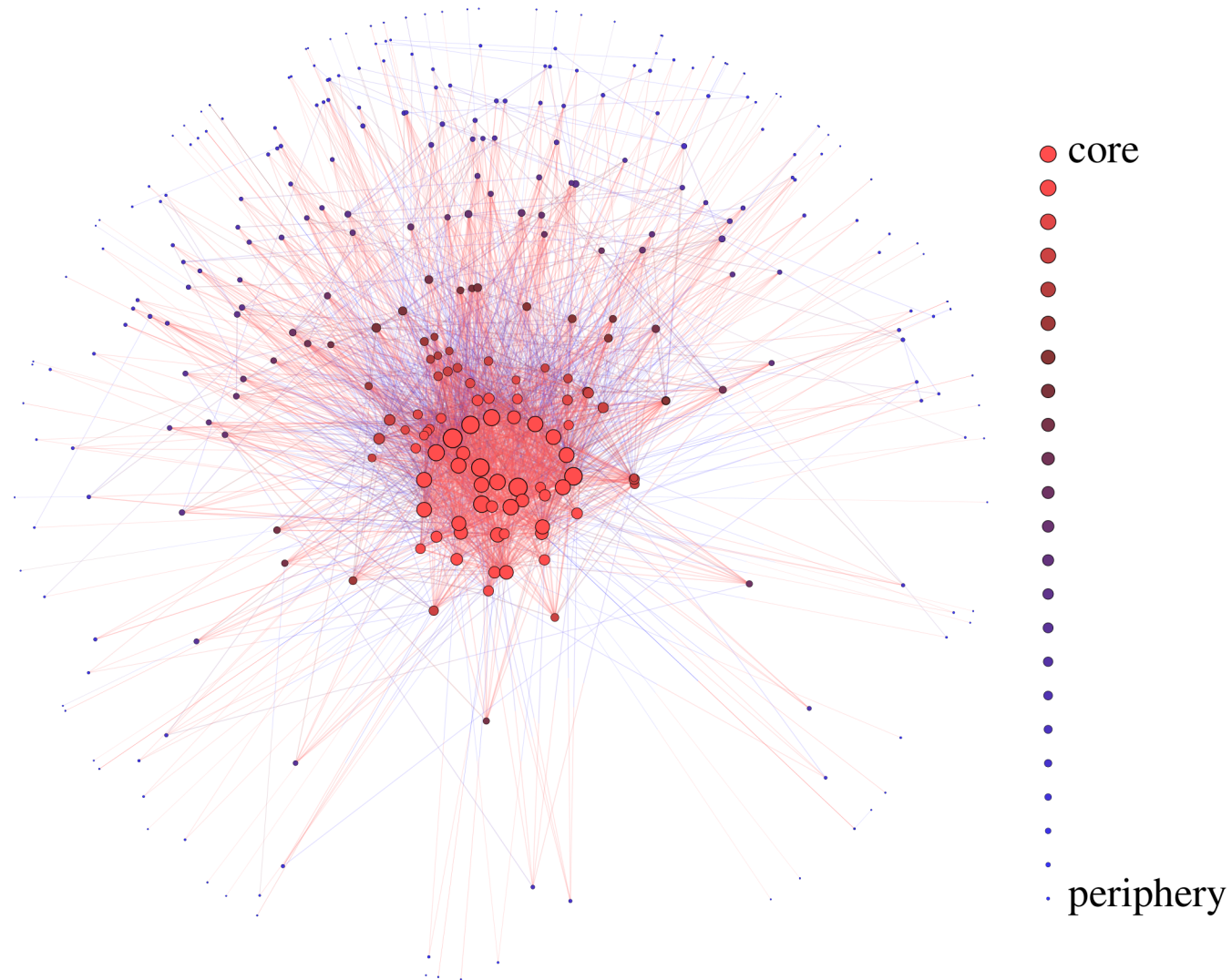
The Figure shows transaction prices and volumes for the second most traded ABS in our sample. While our dataset covers the period from May 16, 2011 to February 29, 2012, the last transaction for the second most traded ABS (which matures after the end of our sample) is a sell to customer on November 16, 2011, where we end the figure. Each panel shows ask prices, interdealer prices and bid prices. Ask prices are for sells to customers and bid prices are for buys from customers.

Figure 4: Dealers' Shares of Customer Volume



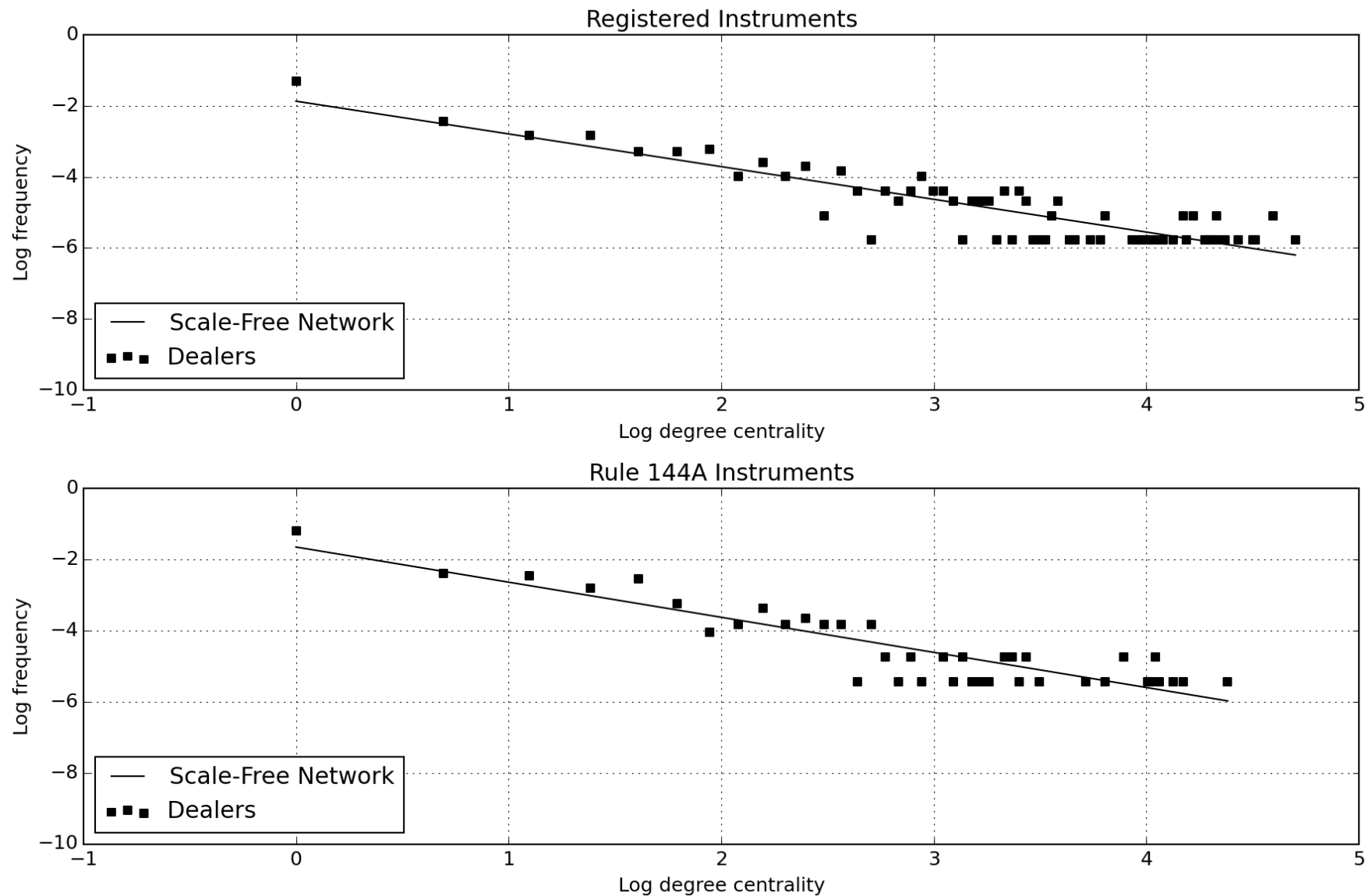
The four panels show Lorenz curves for the 25% of dealers with the largest volumes of original balance traded with customers. All customer trades in the sample period from May 16, 2011 to February 29, 2012 are used to construct Lorenz curves.

Figure 5: The Interdealer Network



The figure shows the interdealer network. Each circle represents a dealer, the size and color of the circle are proportional to the degree centrality—the number of connections the dealer has on the interdealer market, depicted as lines connecting the circles. The figure only includes trades above \$5 million in original balance.

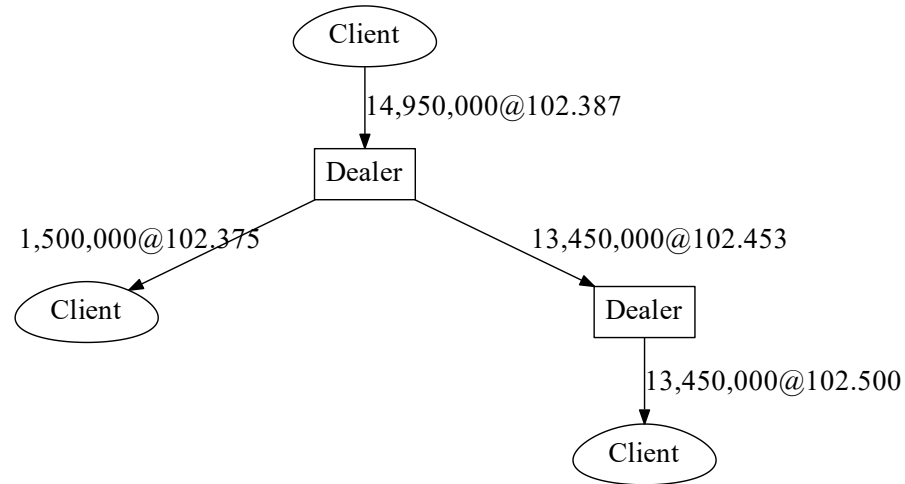
Figure 6: The Interdealer Network Degree Distributions



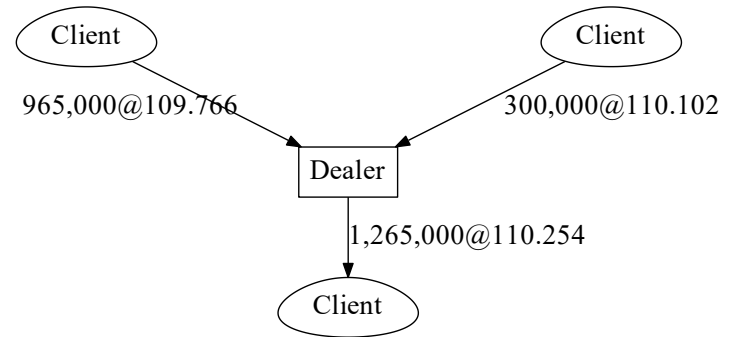
The figure shows the degree distribution for the interdealer network. The x-axis shows the logarithm of degree centrality of a dealer and the y-axis shows the logarithm of its frequency in the distribution among dealers. The scatter plot shows the observations of dealers in the interdealer network, and the line shows the fit of a linear regression. Dealer connections are defined to have at least \$10M of original balance traded between the two dealers in the sample period from May 16, 2011 to February 29, 2012.

Figure 7: Intermediation Trees

Panel A: Intermediation Tree with Two Dealers and a Volume Split



Panel B: Reverse Request with One Dealer



The figure shows two examples of intermediation trees for the most frequently traded Registered ABS instrument in our sample. Each node is a customer seller, a dealer or a customer buyer. Each edge is a trade, labeled by the volume of trade in dollars of original balance and the transaction price. We call an intermediation tree complete when every dollar of the original balance reaches a final customer.

Table 1: Descriptive Statistics for Traded Instruments

	Registered	Rule 144A	Registered ABS	Registered CMBS	Registered CMO	ABS	Rule 144A CMBS	Rule 144A CMO	CDO
Population of CUSIPs	69,231	38,768	4,381	5,603	59,247	7,703	7,720	15,945	7,400
% of CUSIPs traded	31.8	22.8	61.0	47.9	28.1	21.8	22.2	19.9	30.6
Traded CUSIPs	21,993	8,829	2,674	2,686	16,633	1,679	1,711	3,175	2,264
Investment grade (%)	22.8	26.3	48.2	42.5	15.5	31.3	39.4	7.6	38.8
Floating coupon (%)	37.3	19.0	44.9	60.7	32.3	24.6	32.0	19.2	4.6
Average vintage (years)	4.1	2.1	3.5	3.7	4.3	1.5	2.7	1.3	3.2
Average volume per trade (\$M)	14.5	22.4	8.1	17.8	15.0	18.7	47.3	17.7	13.0
Average volume per interdealer trade (\$M)	6.3	19.1	5.1	7.3	6.4	9.2	60.2	10.4	10.3
Average number of trades	15.8	6.5	19.4	21.8	14.3	11.6	7.9	4.3	4.5
Average number of interdealer trades	3.9	1.2	4.1	4.7	3.7	2.7	1.3	0.7	0.9
Smaller-size trades (%)	15.3	2.6	9.5	9.1	17.2	3.3	3.0	3.4	0.7
Number of active dealers per CUSIP	4.2	2.4	5.6	5.9	3.7	3.4	2.6	1.8	2.2

The population of CUSIPs is all unique CUSIPs in the FINRA database up to February 29, 2012. We define traded CUSIPs as having at least one trade over the sample period from May 16, 2011 to February 29, 2012 (208 trading days). Investment grade is based on ratings database provided by FINRA and Moody's website. We classify unrated instruments as high yield rather than investment grade instruments. The average vintage is computed using the first date the security was rated according to the Moody's website. Average volume per trade is the average trade size in the original balance traded averaged across all traded instruments, the average volume per interdealer trade is the average interdealer trade size average across all traded instruments with non-zero interdealer volume. Average number of trades is the number of trade reports averaged across all traded instruments, average number of interdealer trades is the number of interdealer trade reports averaged across all traded instruments with non-zero interdealer volume. Small-Size trades is the number of trades of less than \$100,000 of original balance with customers divided by the number of customer trade reports for each traded instrument and averaged across all traded instruments. Number of active dealers per CUSIP is the number of dealers who participated in at least one trade either with a customer or another dealer over the sample period for each traded instrument and averaged across all traded instruments.

Table 2: Customer Bid-Ask Spreads by Trade Sizes

	Registered	Rule 144A	Diff. in av. Reg. and 144A	ABS	Registered CMBS	CMO	ABS	Rule 144A CMBS	CMO	CDO
Average bid-ask Spread	1.968 (0.007)	0.742 (0.015)	1.226 (0.017)	0.481 (0.011)	0.527 (0.010)	2.859 (0.010)	0.437 (0.021)	0.720 (0.030)	1.022 (0.036)	0.970 (0.039)
Smaller-size spread Average	3.320 (0.013)	1.610 (0.116)	1.709 (0.117)	1.295 (0.036)	1.413 (0.038)	3.676 (0.014)	0.706 (0.115)	1.842 (0.316)	2.224 (0.201)	2.543 (0.574)
Larger-size spread Average	1.191 (0.008)	0.702 (0.015)	0.489 (0.017)	0.327 (0.011)	0.385 (0.010)	2.051 (0.013)	0.423 (0.021)	0.678 (0.029)	0.929 (0.035)	0.946 (0.039)
Diff. in av. smaller and larger	2.129 (0.015)	0.909 (0.117)		0.969 (0.038)	1.028 (0.040)	1.625 (0.019)	0.283 (0.117)	1.164 (0.319)	1.295 (0.205)	1.596 (0.581)

Customer spreads are total realized spreads over an intermediation chain from a client-seller to a client-buyer measured in percentages. We compute smaller-size spreads using intermediation chains with no more than \$100,000 of original par sold to a client-buyer at the end of the chain. Standard errors are reported in parentheses. The sample is from May 16, 2011 to February 29, 2012.

Table 3: Intermediation Chains

	Registered	Rule 144A	ABS	Registered CMBS	CMO	ABS	Rule 144A CMBS	CMO	CDO
Client-sell volume (\$M)	797,675	391,094	93,212	164,088	540,375	82,865	181,832	74,293	52,105
Complete chains	77,045	12,493	13,607	16,799	46,639	4,013	3,363	2,684	2,433
% of volume	83.7	71.0	86.5	78.4	84.9	81.5	66.9	66.1	75.6
One round	56,009	10,651	10,503	13,768	31,738	3,209	2,965	2,251	2,226
Chains w/split volume (%)	11.0	9.2	9.0	13.6	10.5	10.4	13.4	6.4	4.6
Chains w/reverse request (%)	6.0	5.7	6.9	7.8	4.9	7.9	7.2	3.1	3.0
Avg. number of clients	2.5	2.3	2.3	2.4	2.6	2.3	2.4	2.3	2.3
More than one round	21,036	1,842	3,104	3,031	14,901	804	398	433	207
Avg. length given > 1 round	2.8	2.6	2.5	2.7	3.0	2.7	2.4	2.4	2.4
Chains w/split volume (%)	14.5	21.2	18.5	21.3	12.2	25.6	21.6	12.7	20.8
Chains w/reverse request (%)	6.3	8.9	8.5	11.8	4.8	9.6	11.8	6.9	4.8
Avg. number of clients	3.7	2.7	2.9	2.9	4.3	2.8	2.7	2.5	2.7

For each category of instruments, we take client sell trades and construct intermediation trees as depicted on Figure 6. For some client sell trades we do not find any matching trade and the volume remains in dealer's inventory during our sample period. We call an intermediation tree complete when we are able to find client-buy trades at the end of the branches of the tree that sum up to the initial client-sell volume. % of volume shows the percentage of the client-sell volume for which we find complete intermediation trees. One round intermediation trees have one dealer intermediating between a client-seller and client-buyers. Trees have split volume when there is more than one client-buyer at the end of the intermediation tree. Average number of dealers given more than one round is the average number of dealers who participated in intermediation trees with more than one round of intermediation.

Table 4: Dealers in Intermediation Chains

	Registered			Rule 144A			Registered			Rule 144A		
	Registered	Rule 144A	ABS	CMBS	CMO	ABS	CMBS	CMO	ABS	CMBS	CMO	CDO
Core dealers	33	23	33	33	33	23	23	23	23	23	23	21
Number of matches	97,762	15,312	16,803	27,168	53,791	5,897	4,244	2,553	2,618			
Avg. holding period (days)	13.9	22.1	13.9	16.6	12.6	22.5	24.6	16.5	22.5			
Vol. w/< 1 day holding period (%)	69.7	69.8	60.4	56.1	76.3	65.0	67.6	76.3	74.9			
Small-size matches (%)	30.2	2.3	11.0	10.4	46.1	2.2	2.9	3.1	1.0			
Matches w/split volume (%)	29.9	26.4	23.5	30.4	31.7	28.1	31.4	21.2	19.9			
Matches w/buy from a client (%)	72.2	84.0	83.7	77.4	66.0	73.9	91.0	89.3	90.3			
w/sell to a dealer (%)	15.2	14.3	16.6	14.3	15.2	19.7	9.6	13.7	12.9			
Matches w/buy from dealer (%)	27.8	16.0	16.3	22.6	34.0	26.1	9.0	10.7	9.7			
w/sell to a dealer (%)	36.9	57.1	33.1	50.9	32.8	61.1	50.5	46.9	53.3			
Peripheral dealers	528	240	242	193	465	148	115	146	112			
Number of matches	63,985	6,034	8,042	5,805	50,138	2,437	1,219	1,513	865			
Avg. holding period (days)	6.3	9.0	5.9	4.5	6.6	12.3	10.1	5.4	4.5			
Vol. w/< 1 day holding period (%)	83.1	86.6	89.6	75.6	83.2	82.0	86.6	91.9	90.1			
Small-size matches (%)	65.5	6.8	34.7	40.4	73.4	5.8	7.7	5.6	10.9			
Matches w/split volume (%)	9.8	14.1	14.0	7.1	9.4	19.7	11.6	9.8	9.5			
Matches w/buy from a client (%)	52.9	65.4	52.4	56.9	52.5	59.9	66.4	66.3	78.2			
w/sell to a dealer (%)	58.9	34.5	55.2	56.3	59.8	38.4	31.6	33.4	31.5			
Matches w/buy from dealer (%)	47.1	34.6	47.6	43.1	47.5	40.1	33.6	33.7	21.8			
w/sell to a dealer (%)	52.0	36.3	39.3	56.0	53.6	41.2	34.0	30.4	31.7			

A match is a pair comprising a buy- and a sell-trade by a dealer. When a dealer sells to more than one party in a chain we construct more than one match. We use eigenvector degree centrality to classify dealers. Core dealers are in the top 5% of the eigenvector centrality distribution among dealers in the interdealer network, computed separately for Registered and Rule 144A instruments, all other dealers are peripheral. Avg. holding period in days is the average number of days in dealer's inventory per match; vol. with less than one day holding period is the volume-weighted share of matches that have less than one-day holding period. Small-size matches show the percentage of matches with no more than \$100,000 volume. Matches with buy from a client are matches in which the dealer buys from a client, all other matches have the dealer buying from another dealer. In matches with split volume the dealer sells the volume to multiple counterparties. Matches with sell to a dealer are matches where the dealer sells to another dealer.

Table 5: Customer Bid-Ask Spreads by Dealer Centrality

		Registered					Rule 144A			
		Registered	Rule 144A	ABS	CMBS	CMO	ABS	CMBS	CMO	CDO
Core dealers										
Larger-size	75 th perc.	0.995	0.385	0.102	0.412	3.304	0.193	0.596	0.454	0.436
	50 th perc.	0.188	0.104	0.027	0.090	0.838	0.057	0.170	0.125	0.130
	25 th perc.	0.023	0.000	0.004	0.007	0.210	0.004	0.011	0.000	0.000
Smaller-size	75 th perc.	3.636	0.509	0.820	0.524	3.656	0.165	0.774	0.530	0.181
	50 th perc.	3.026	0.148	0.154	0.121	3.468	0.086	0.332	0.190	0.131
	25 th perc.	0.644	0.039	0.027	0.019	2.597	0.015	0.050	0.062	0.083
Peripheral dealers										
Larger-size	75 th perc.	1.993	1.187	0.736	0.635	3.317	0.612	1.169	1.864	1.742
	50 th perc.	0.553	0.333	0.277	0.212	1.212	0.185	0.343	0.493	0.595
	25 th perc.	0.124	0.063	0.035	0.052	0.266	0.023	0.037	0.125	0.184
Smaller-size	75 th perc.	4.608	2.857	2.421	2.599	4.853	0.703	3.604	3.509	4.100
	50 th perc.	2.943	0.548	0.912	0.870	3.085	0.259	0.995	0.983	2.503
	25 th perc.	1.731	0.111	0.240	0.251	2.017	0.000	0.128	0.156	1.499

Customer spreads are total realized spreads from a client-seller to a client-buyer in an intermediation chain measured in percentages. We compute smaller-size spreads using intermediation chains with no more than \$100,000 of original par sold to a client-buyer at the end of the chain. We use eigenvector degree centrality to classify dealers. Core dealers are in the top 5% of the eigenvector centrality distribution among dealers in the interdealer network, computed separately for Registered and Rule 144A instruments, all other dealers are peripheral. The sample is from May 16, 2011 to February 29, 2012. The 75th percentile, median, and the 25th percentile spreads are reported.

Table 6: Definitions of Variables used in Regressions

Bond Controls	4-6 years Vintage Dummy	Dummy variable equal to one when the time between the issuance date and the trade is between four and six years. The issuance date is the first coupon date or the first Moody's rating date.
	> 6 years Vintage Dummy	Dummy variable equal to one when the time between the issuance date and the trade is greater than six years. The issuance date is the first coupon date or the first Moody's rating date.
	Investment Grade Dummy	Dummy variable equal to one when the credit rating is above BBB level based either on Moody's rating or the credit rating provided by the FINRA bond dataset.
	Floating Coupon Dummy	Dummy variable equal to one when the CUSIP has floating coupon type in the FINRA bond dataset.
	Number of Trades in CUSIP	The number of trades per trading day in the CUSIP during the sample period, which includes both customers and interdealer trades.
	Active Dealers in CUSIP	The number of dealers who traded the given CUSIP with customers or with other dealers over the sample period.
	Average Match Volume per CUSIP	The logarithm of the original balance volume traded in the intermediation chain, averaged across all intermediation chains for the given CUSIP.
	Deviation of Trade Volume from Average	The difference between the logarithm of the original balance volume traded in the intermediation chain and the Average Match Volume per CUSIP defined above.
	Smaller-Size Trade Dummy	Dummy variable equal to one when the trade volume is below \$100,000 original balance.
	Split Dummy	Dummy variable equal to one when the original balance volume sold to a counterparty by the dealer is different from the original balance volume bought from a counterparty, e.g. as shown in Figure 7.
	Dealer (First in Chain) Centrality Dummy	Dummy variable equal to one when the eigenvector degree centrality of a dealer in the interdealer network, or the average eigenvector degree centrality of the dealers participating in an intermediation chain, is in the top 5% of the eigenvector centrality distribution among dealers, and zero otherwise.
	Core Dealer Later in Chain	Dummy variable equal to one when the eigenvector degree centrality is in the top 5% of the eigenvector centrality distribution among dealers in the interdealer network, zero otherwise.
	Chain Length	Number of trading rounds in the intermediation chain.
	Time to Complete Chain	The time in days between a buy trade and a sell trade in the match used to compute the bid-ask spread.
	Buy from Customer Dummy	Dummy variable equal to one when a dealer buys from a customer and sells to another dealer.
	Sell to Customer Dummy	Dummy variable equal to one when a dealer buys from another dealer and sells to a customer.
Market Controls	Mortgage Rate Changes	Changes from bid to ask trade in the mortgage rate (principal component of the 1-year, 15-year, and 30-year mortgage rate).
	Credit Spread Changes	Changes from bid to ask trade in the Moody's corporate bond Baa spread over 10-year Treasury rate.
	TED Spread Changes	Changes from bid to ask trade in the TED spread.
	Term Structure Slope Changes	Changes from bid to ask trade in the slope of the Treasury term structure (3-month to 10-year).

Table 7: Customer Time to Complete and Chain Lengths Regressions

	Full Sample				Most Traded			
	Registered		Rule 144A		Registered		Rule 144A	
	(1)	(2)	(1)	(2)	(1)	(2)	(1)	(2)
	Time to Complete	Chain Length	Time to Complete	Chain Length	Time to Complete	Chain Length	Time to Complete	Chain Length
4-6 Years Vintage Dummy	-0.057 (0.012)	0.023 (0.009)	-0.233 (0.028)	-0.050 (0.027)	-0.203 (0.057)	-0.105 (0.041)	-0.555 (0.321)	0.059 (0.243)
> 6 Years Vintage Dummy	-0.150 (0.013)	0.061 (0.011)	-0.240 (0.027)	0.012 (0.031)	-0.217 (0.062)	0.119 (0.061)	-0.548 (0.271)	-0.261 (0.366)
Investment Grade Dummy	0.018 (0.009)	-0.015 (0.010)	0.130 (0.021)	-0.014 (0.023)	-0.236 (0.086)	-0.284 (0.101)	0.093 (0.127)	-0.219 (0.226)
Floating Coupon Dummy	-0.263 (0.010)	-0.175 (0.009)	-0.099 (0.027)	0.062 (0.031)	0.048 (0.051)	0.125 (0.043)	0.226 (0.237)	0.234 (0.257)
Number of Trades in CUSIP	0.039 (0.006)	0.048 (0.008)	-0.001 (0.035)	-0.121 (0.042)	0.100 (0.022)	-0.007 (0.016)	0.239 (0.232)	0.057 (0.235)
Active Dealers in CUSIP	0.050 (0.006)	0.115 (0.007)	0.129 (0.026)	0.182 (0.033)	-0.109 (0.037)	0.104 (0.034)	-0.006 (0.238)	-0.031 (0.242)
Dealer (First in Chain) Centrality Dummy	0.146 (0.009)	-0.347 (0.008)	0.183 (0.024)	-0.338 (0.028)	0.060 (0.038)	-0.711 (0.046)	0.006 (0.104)	-0.250 (0.157)
R-sq. adj.	0.201	0.124	0.201	0.124	0.225	0.220	0.225	0.220
Collateral type fixed effects			Yes				Yes	
Market controls			Yes				Yes	
Observations			76,775				4,267	

We use eigenvector degree centrality in the regression. Most traded instruments are the top 50 Registered instruments, and the top 25 Rule 144A instruments in terms of the number of customer spread observations, excluding those with more than 90% of spreads being smaller-size or larger-size). The regression includes fixed-effects for each collateral type, and market control variables (changes from bid to ask trade): Moody's Baa credit spread, slope of the Treasury term structure, TED spread, and mortgage rate. Standard errors are reported in parentheses.

Table 8: Customer Bid-Ask Spreads Regression

	Full Sample				Most Traded			
	Registered		Rule 144A		Registered		Rule 144A	
Number of Trades in CUSIP	0.094 (0.011)	0.101 (0.010)	0.016 (0.037)	0.029 (0.035)	0.053 (0.027)	0.053 (0.027)	0.228 (0.276)	0.109 (0.278)
Active Dealers in CUSIP	0.014 (0.012)	0.012 (0.012)	-0.151 (0.034)	-0.156 (0.033)	0.054 (0.046)	0.051 (0.045)	-0.018 (0.240)	0.038 (0.251)
Unexpected Chain Length	1.067 (0.018)	0.827 (0.030)	0.515 (0.044)	0.394 (0.038)	0.735 (0.053)	0.310 (0.087)	0.666 (0.192)	0.388 (0.091)
Unexpected Time to Complete Chain	0.274 (0.012)	0.232 (0.015)	0.094 (0.025)	0.098 (0.025)	0.074 (0.030)	0.153 (0.034)	0.107 (0.046)	-0.027 (0.039)
Smaller-Size Trade Dummy	0.775 (0.034)	0.780 (0.035)	0.493 (0.127)	0.649 (0.199)	0.191 (0.083)	0.103 (0.083)	0.504 (0.227)	1.122 (0.457)
Average Match Volume per CUSIP	-0.340 (0.021)	-0.379 (0.021)	-0.021 (0.025)	-0.006 (0.025)	-0.616 (0.058)	-0.642 (0.059)	-0.296 (0.116)	-0.176 (0.122)
Deviation of Trade Volume from Average	-0.048 (0.009)	-0.123 (0.010)	-0.000 (0.014)	-0.023 (0.013)	-0.265 (0.029)	-0.185 (0.028)	0.025 (0.085)	-0.146 (0.072)
Dealer (First in Chain) Centrality Dummy	-0.410 (0.022)	-0.418 (0.031)	-0.635 (0.052)	-0.638 (0.052)	-0.575 (0.061)	-0.346 (0.079)	-0.293 (0.154)	-0.183 (0.122)
Core Dealer Later in Chain	-0.229 (0.054)	-0.183 (0.055)	-1.489 (0.203)	-1.140 (0.188)	-0.210 (0.181)	0.080 (0.184)	-0.768 (0.590)	-0.937 (0.638)
Smaller-Size Trade * Deviation of Trade Volume	- (0.018)	0.152 (0.018)	- (0.096)	0.201 (0.096)	- (0.052)	-0.171 (0.052)	- (0.332)	0.721 (0.332)
Smaller-Size Trade * Dealer Centrality	- (0.046)	0.141 (0.046)	- (0.256)	0.252 (0.256)	- (0.125)	-0.163 (0.125)	- (0.501)	-0.126 (0.501)
Deviation of Trade Volume * Dealer Centrality	- (0.017)	0.145 (0.017)	- (0.038)	0.035 (0.038)	- (0.052)	0.236 (0.052)	- (0.186)	0.047 (0.186)
Smaller-Size Trade * Unexpected Length	- (0.031)	0.330 (0.031)	- (0.151)	0.909 (0.151)	- (0.088)	0.531 (0.088)	- (0.296)	0.499 (0.296)
Smaller-Size Trade * Unexpected Time	- (0.006)	0.021 (0.006)	- (0.030)	0.028 (0.030)	- (0.012)	-0.030 (0.012)	- (0.037)	0.097 (0.037)
Collateral type fixed effects	Yes		Yes		Yes		Yes	
Bond and market controls	Yes		Yes		Yes		Yes	
Observations	76,775		76,775		4,267		4,267	
R-sq.	0.622	0.626	0.622	0.626	0.828	0.834	0.828	0.834

The regression includes fixed-effects for each collateral type, bond control variables: 4-6 years vintage dummy, > 6 years vintage dummy, investment grade dummy, floating coupon dummy, and market control variables (changes from bid to ask trade): Moody's Baa credit spread, slope of the Treasury term structure, TED spread, and mortgage rate. Standard errors are reported in parentheses. Standard errors are clustered within trade settlement dates, instrument subcategory and placement type.

Table 9: Interdealer Bid-Ask Spreads Regression

	Full Sample				Most Traded			
	Registered		Rule 144A		Registered		Rule 144A	
Sell to	0.158	0.163	-0.144	-0.117	0.349	0.358	-1.562	-1.466
Dummy	(0.032)	(0.032)	(0.097)	(0.096)	(0.177)	(0.176)	(0.806)	(0.771)
Buy from	-0.584	-0.552	-0.289	-0.271	-0.624	-0.565	-1.760	-1.677
Dummy	(0.052)	(0.050)	(0.105)	(0.101)	(0.181)	(0.182)	(0.878)	(0.862)
Split	0.241	0.244	0.057	0.066	0.164	0.145	0.257	0.210
Dummy	(0.033)	(0.032)	(0.076)	(0.076)	(0.112)	(0.111)	(0.312)	(0.345)
Dealer	0.053	0.063	-0.256	-0.236	-0.078	0.088	-0.011	0.317
Centrality Dummy	(0.039)	(0.069)	(0.089)	(0.086)	(0.136)	(0.189)	(0.318)	(0.343)
Chain	-0.097	-0.075	-0.047	-0.062	-0.116	-0.164	-0.373	-0.526
Length	(0.013)	(0.020)	(0.026)	(0.027)	(0.075)	(0.156)	(0.212)	(0.247)
Time	0.271	0.210	0.204	0.169	0.237	0.059	-0.125	0.001
to Complete Chain	(0.023)	(0.050)	(0.048)	(0.043)	(0.067)	(0.096)	(0.158)	(0.114)
Smaller-Size Trade	0.192	0.196	0.587	1.018	0.116	0.059	-0.296	-0.061
Dummy	(0.042)	(0.043)	(0.261)	(0.359)	(0.206)	(0.210)	(0.652)	(0.751)
Average Match Volume	-0.241	-0.248	-0.266	-0.253	-0.221	-0.277	-0.213	-0.172
per CUSIP	(0.031)	(0.030)	(0.057)	(0.058)	(0.097)	(0.097)	(0.280)	(0.295)
Deviation of Trade	-0.100	-0.083	-0.045	-0.062	-0.390	-0.390	-0.126	-0.314
Volume from Average	(0.024)	(0.040)	(0.024)	(0.021)	(0.081)	(0.092)	(0.158)	(0.126)
Smaller-Size Trade *		-0.015		0.249		-0.043		0.362
Deviation of Trade Volume		(0.035)		(0.131)		(0.135)		(0.348)
Smaller-Size Trade *		0.024		-0.340		-0.299		-0.853
Dealer Centrality		(0.071)		(0.516)		(0.313)		(0.611)
Deviation of Trade Volume *		-0.039		0.093		-0.112		-0.616
Dealer Centrality		(0.035)		(0.047)		(0.133)		(0.225)
Smaller-Size Trade *		-0.028		0.765		0.092		0.147
Chain Length		(0.022)		(0.271)		(0.173)		(0.256)
Smaller-Size Trade *		0.108		0.752		0.342		-0.268
Time to Complete		(0.054)		(0.482)		(0.117)		(0.324)
Collateral type fixed effects	Yes		Yes		Yes		Yes	
Bond and market controls	Yes		Yes		Yes		Yes	
Observations	65,409		65,409		2,503		2,503	
R-sq.	0.537	0.538	0.537	0.538	0.680	0.686	0.680	0.686

The regression includes fixed-effects for each collateral type, bond control variables: 4-6 years vintage dummy, > 6 years vintage dummy, investment grade dummy, floating coupon dummy, and market control variables (changes from bid to ask trade): Moody's Baa credit spread, slope of the Treasury term structure, TED spread, and mortgage rate. Standard errors are reported in parentheses. Standard errors are clustered within trade settlement dates, instrument subcategory and placement type.

Table 10: Customer Bid-Ask Spreads Quantile Regression

	Full sample		Most traded	
	Registered	Rule 144A	Registered	Rule 144A
Core dealers				
Mean (%)	1.218	0.474	2.068	0.292
St. dev. (%)	2.147	1.764	1.881	1.097
Observations	32,729	6,297	2,687	136
Peripheral dealers				
Mean (%)	2.604	1.264	2.304	1.497
St. dev. (%)	3.092	2.598	2.707	2.452
Observations	37,065	3,368	1,391	138
Quantile regressions				
Coefficient estimates on dealer centrality				
0.10	-0.093 (0.006)	-0.010 (0.007)	-0.018 (0.053)	-0.076 (0.028)
0.25	-0.057 (0.006)	-0.028 (0.002)	-0.042 (0.015)	-0.101 (0.028)
0.50	-0.021 (0.008)	-0.138 (0.006)	-0.124 (0.015)	-0.292 (0.091)
0.75	-0.471 (0.013)	-0.441 (0.023)	-0.614 (0.030)	-0.551 (0.077)
0.90	-1.223 (0.031)	-0.733 (0.062)	-1.095 (0.078)	-1.296 (0.427)
Category fixed effects	Yes	Yes	Yes	Yes
Bond and market controls	Yes	Yes	Yes	Yes
Observations	69,794	9,665	4,078	274

We use eigenvector degree centrality in the quantile regressions to classify dealers. Core dealers are in the top 5% of the eigenvector centrality distribution among dealers in the interdealer network, others are peripheral. The most traded instruments are the top 50 Registered instruments and the top 25 Rule 144A instruments in terms of the number of customer spread observations, excluding instruments with less than 10% of their spreads smaller-size or less than 10% of their spreads larger-size. The quantile regression includes fixed-effects for ABS, CMBS, CMO, and CDO categories, bond controls: 4-6 years vintage dummy, > 6 years vintage dummy, investment grade dummy, floating coupon dummy, and market controls (changes from bid to ask trade): Moody's Baa credit spread, slope of the Treasury term structure, TED spread, and mortgage rate. Standard errors are reported in parentheses.

Table 11: VAR Analysis of the Variance of Long-Term Returns

Panel A: Relative contributions to the variance of long-term returns when the ordering of the innovations is: signed log-volume, dealer centrality dummy, log-return

	Most Traded		Second-Most Traded	
	Registered	Rule 144A	Registered	Rule 144A
Explained by signed log-volume (%)	12.1 (0.8)	2.8 (1.7)	14.2 (1.2)	0.1 (3.3)
Explained by dealer centrality dummy (%)	0.7 (0.2)	2.4 (1.4)	0.9 (0.3)	1.0 (3.4)
Explained by log-return (%)	87.2 (0.8)	94.8 (2.8)	84.9 (1.2)	98.9 (5.2)

Panel B: Relative contributions to the variance of long-term returns when the ordering of the innovations is: dealer centrality dummy, signed log-volume, log-return

	Most Traded		Second-Most Traded	
	Registered	Rule 144A	Registered	Rule 144A
Explained by signed log-volume (%)	6.1 (0.6)	4.9 (2.5)	5.0 (0.8)	1.0 (3.9)
Explained by dealer centrality dummy (%)	6.7 (0.6)	0.3 (0.5)	10.2 (1.0)	0.1 (3.2)
Explained by log-return (%)	87.2 (0.9)	94.8 (2.7)	84.9 (1.2)	98.9 (5.8)

We use eigenvector degree centrality in the quantile regressions to classify dealers. Core dealers are in the top 5% of the eigenvector centrality distribution among dealers in the interdealer network, others are peripheral. We sort instruments in terms of the number of customer spread observations, excluding instruments with less than 10% of their spreads smaller-size or less than 10% of their spreads larger-size. We pick the most traded instruments as the top 50 Registered instruments and the top 25 Rule 144A instruments in that sorting, the next 50 Registered instruments and 25 Rule 144A instruments are the second-most traded instruments. We estimate a panel VAR in equation (7) to calculate the relative contributions to the variance of long-term returns under two alternative variable ordering. Standard errors based on 100 Monte Carlo simulations are reported in parentheses.